

Document

Project: DevOps Implementation

Note: Contents have been deleted. This is not a full version of the document.

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Jean-Philippe LO SIOU

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DOCUMENT CONTROL

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11/03/2025	Jean-Philippe LO SIOU	0.1	First Drafts
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OBJECTIVE

This project consists of hands-on experience in DevOps practices, implementing automation, and creating a fully integrated CI/CD pipeline. By the end, this project is a fully automated system where changes are automatically tested, deployed, and maintained.

Expected Deliverables

1. A **GitHub repository** containing the forked project with all DevOps implementations.
2. A **fully automated CI/CD pipeline**.
3. An **Infrastructure as Code (IaC) setup**.
4. A **containerized application deployed to the cloud**.
5. A **detailed report explaining the execution, challenges faced, and learnings**.

SELECTION & FORKING A REPOSITORY

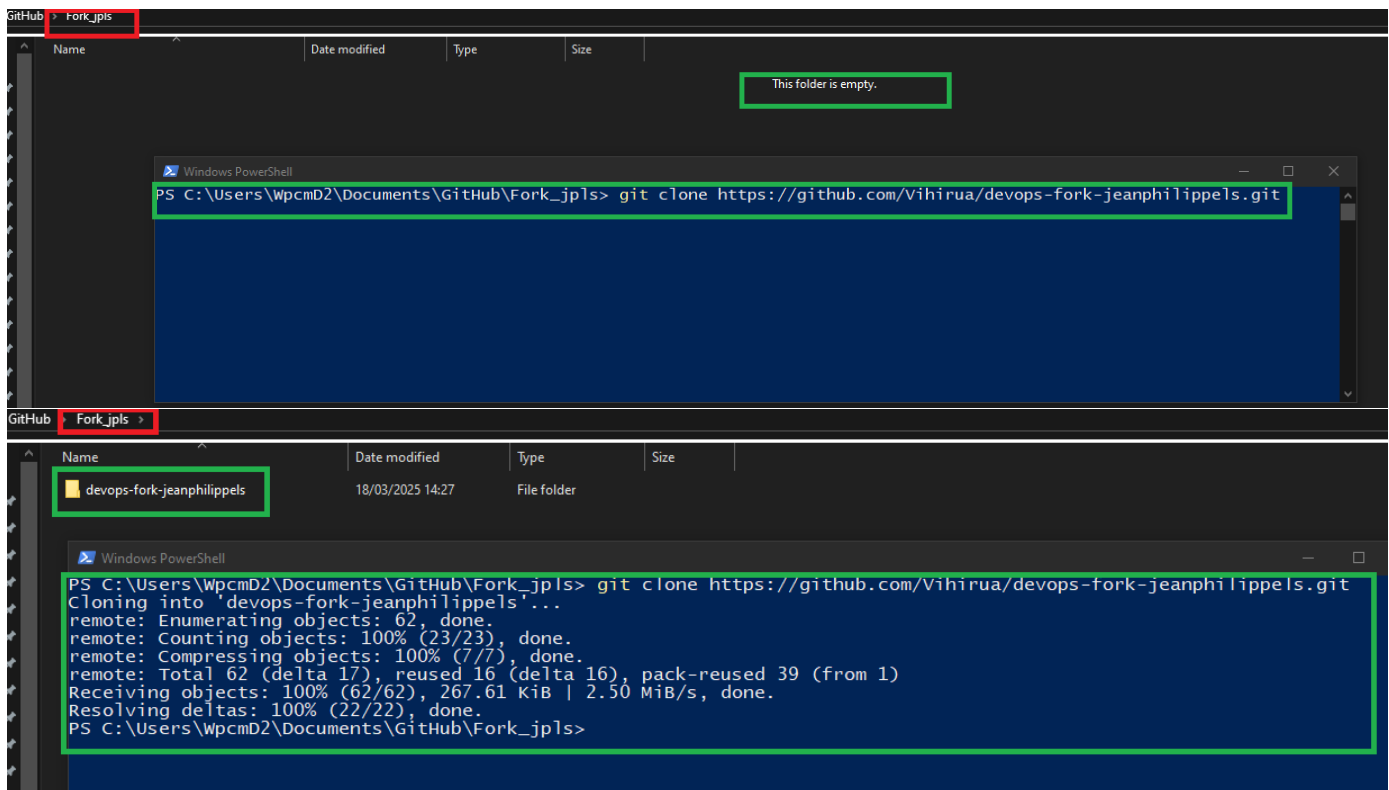
Git & Version Control

- **Forking a project**
- **Create a new branch for DevOps changes**
- **Practice Git Workflow:**
 - **Make changes and commit**
 - **Push changes to GitHub.**
 - **Create a Pull Request (PR) and merge changes.**

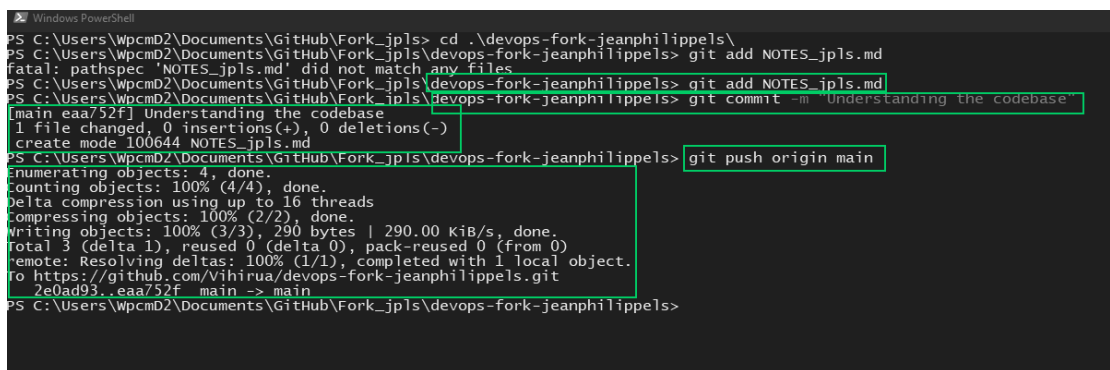
- **Forking a project**

The screenshot shows the GitHub interface for creating a new fork. At the top, the repository path 'felixcameron17 / project-setup-environments' is highlighted with a red box. Below the repository name, a red box contains the text 'Create a new fork'. The form includes fields for 'Owner' (Vihirua) and 'Repository name' (devops-fork-jeanphilippel), which is highlighted with a green box. A green checkmark indicates the name is available. Below the form, there is a checkbox for 'Copy the main branch only' which is checked. At the bottom right, a green box highlights the 'Create fork' button.

The screenshot shows the GitHub repository page for 'devops-fork-jeanphilippels' by user 'Vihirua'. The repository name is highlighted with a green box. Below the repository name, it says 'forked from felixcameron17/project-setup-environments'. The page shows the repository is up to date with the upstream repository. Below this, there is a list of files and their commit history. The files listed are: .github/workflows, MainPage.js, README.md, banner.jpg, banner2.jpg, bootstrap-grid.min.css, bootstrap-reboot.min.css, and bootstrap.min.css. The commit history for each file shows the commit message, the user who made the commit, and the time ago. The commit history for 'MainPage.js' shows 'Update MainPage.js' by 'felixcameron17' 14 commits ago. The commit history for 'bootstrap.min.css' shows 'Add files via upload' by 'felixcameron17' 3 years ago.



- **Git clone**
- **Create a new branch for DevOps changes**



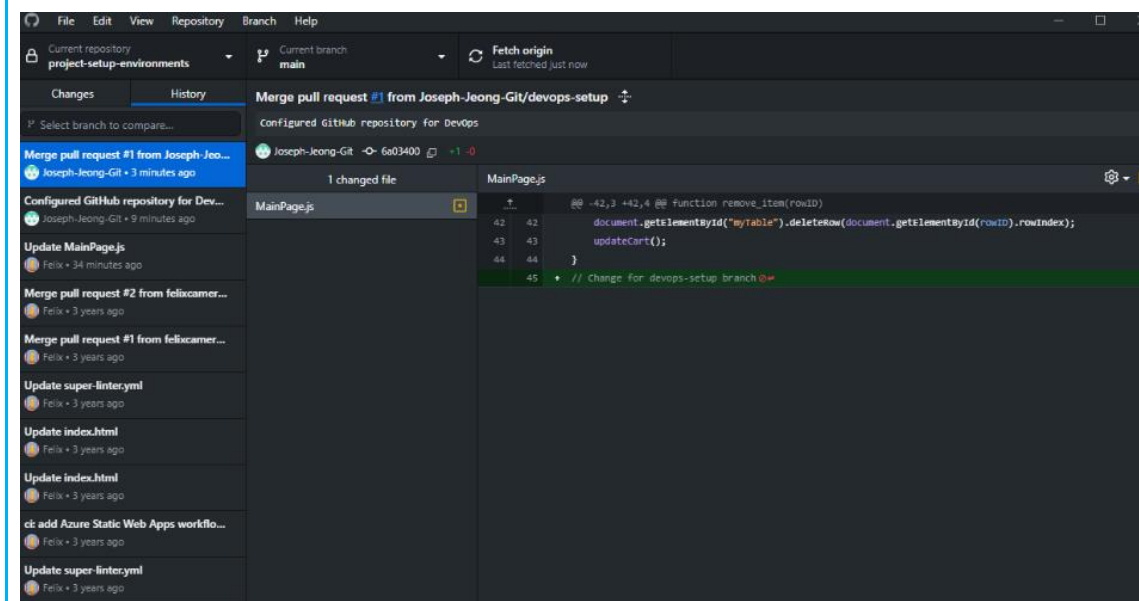
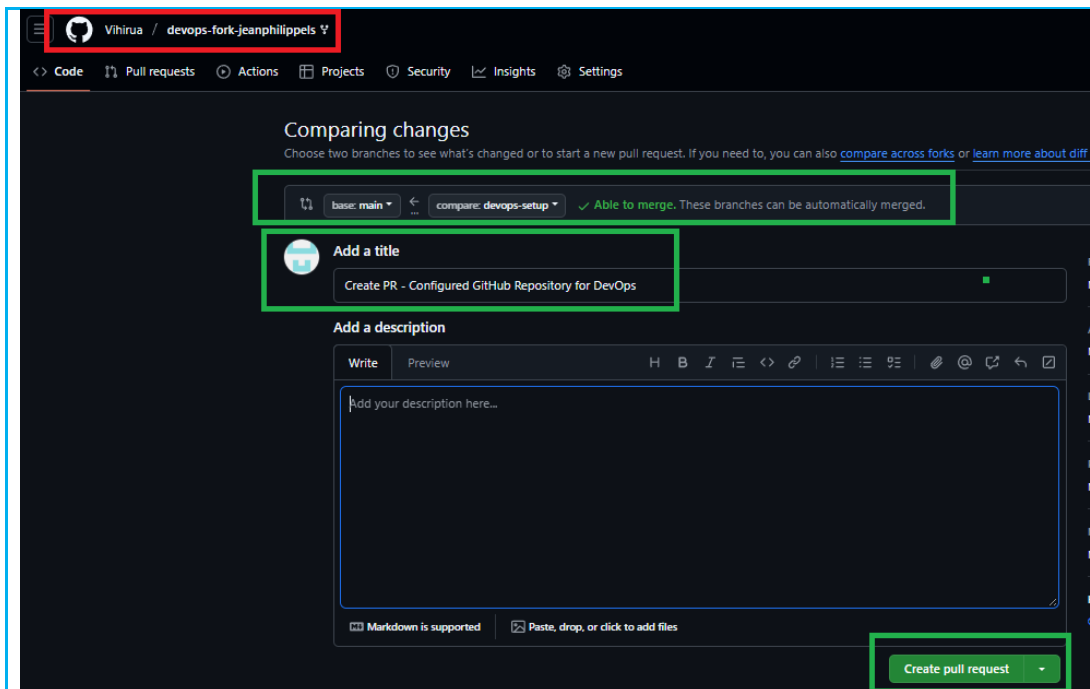
- **Git add NOTES_jpls.md**
- **Git commit -m "Understanding the codebase"**

- **Git push origin main**

- **Push changes to GitHub.**

```
Windows PowerShell
PS C:\Users\Wpcmd2\Documents\GitHub\Fork_jpls> cd .\devops-fork-jeanphilippels\
PS C:\Users\Wpcmd2\Documents\GitHub\Fork_jpls\devops-fork-jeanphilippels> git add NOTES_jpls.md
fatal: pathspec 'NOTES_jpls.md' did not match any files
PS C:\Users\Wpcmd2\Documents\GitHub\Fork_jpls\devops-fork-jeanphilippels> git add NOTES_jpls.md
PS C:\Users\Wpcmd2\Documents\GitHub\Fork_jpls\devops-fork-jeanphilippels> git commit -m "Understanding the codebase"
[main eaa752f] Understanding the codebase
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 NOTES_jpls.md
PS C:\Users\Wpcmd2\Documents\GitHub\Fork_jpls\devops-fork-jeanphilippels> git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 290 bytes | 290.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Vihirua/devops-fork-jeanphilippels.git
 2e0ad93..eaa752f  main -> main
PS C:\Users\Wpcmd2\Documents\GitHub\Fork_jpls\devops-fork-jeanphilippels>
```

- **Create a Pull Request (PR) and merge changes.**



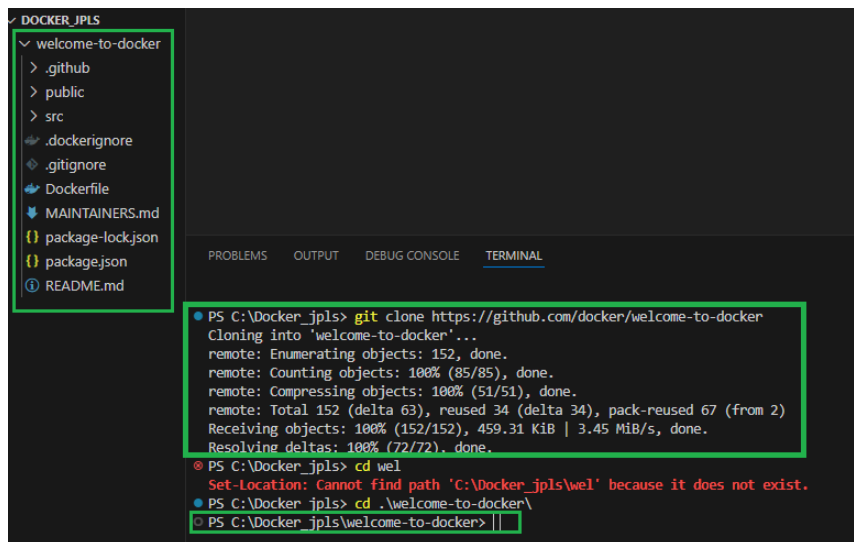
CONTAINERIZATION WITH DOCKER & KUBERNETES

Deployment in Local Docker Container

- **Run a Docker container**
- **Push Docker Image to Docker Hub.**
- **Building DevOps image with Dockerfile**
- **Building container with the DevOps image**
- **Testing container with the DevOps image**
- **Get into the DevOps docker container**
- **Pushing DevOps image to private docker repository**

Docker is a software allowing to build, test and deploy applications by using containers. It packages everything in containers that the applications need to run with (Libraries, systems tools, code). It can quickly deploy and scale applications into any environments (any operating Systems)

- **Run a Docker container**



```
PS C:\Docker_jpls> git clone https://github.com/docker/welcome-to-docker
Cloning into 'welcome-to-docker'...
remote: Enumerating objects: 152, done.
remote: Counting objects: 100% (85/85), done.
remote: Compressing objects: 100% (51/51), done.
remote: Total 152 (delta 63), reused 34 (delta 34), pack-reused 67 (from 2)
Receiving objects: 100% (152/152), 459.31 KiB | 3.45 MiB/s, done.
Resolving deltas: 100% (72/72), done.
PS C:\Docker_jpls> cd wel
Set-Location: Cannot find path 'C:\Docker_jpls\wel' because it does not exist.
PS C:\Docker_jpls> cd .\welcome-to-docker\
PS C:\Docker_jpls\welcome-to-docker> |
```

- **Git clone**

The Dockerfile :

```

1 # Start your image with a node base image
2 FROM node:18-alpine
3
4 # The /app directory should act as the main application directory
5 WORKDIR /app
6
7 # Copy the app package and package-lock.json file
8 COPY package*.json ./
9
10 # Copy local directories to the current local directory of our docker image (/app)
11 COPY ./src ./src
12 COPY ./public ./public
13
14 # Install node packages, install serve, build the app, and remove dependencies at the end
15 RUN npm install \
16     && npm install -g serve \
17     && npm run build \
18     && rm -fr node_modules
19
20 EXPOSE 3000
21
22 # Start the app using serve command
23 CMD [ "serve", "-s", "build" ]

```

```

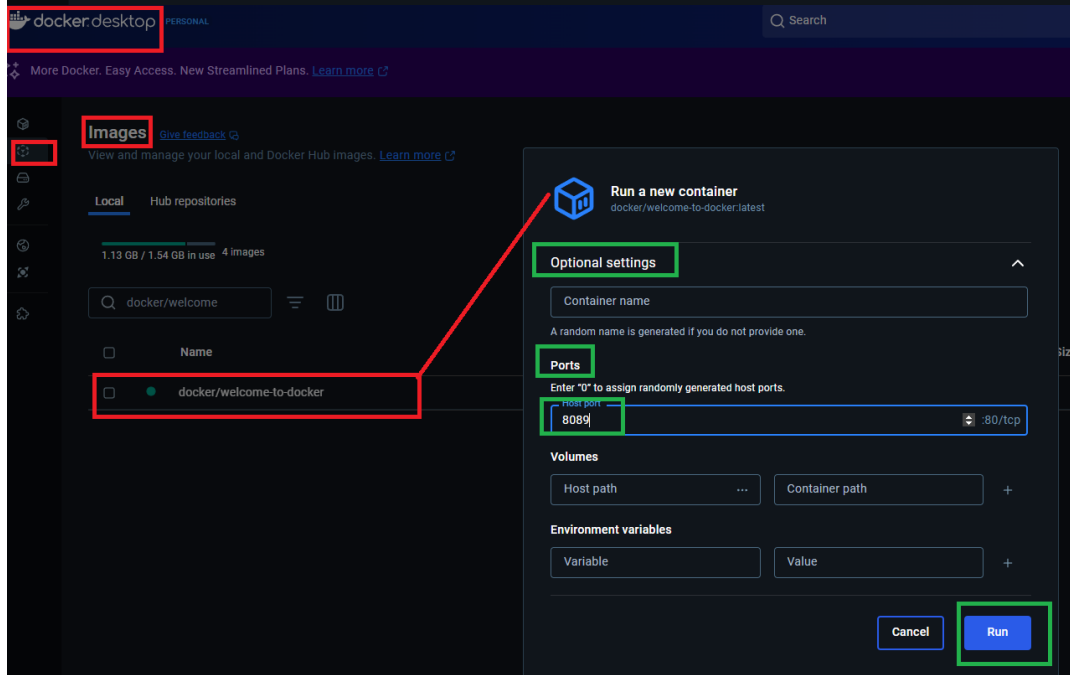
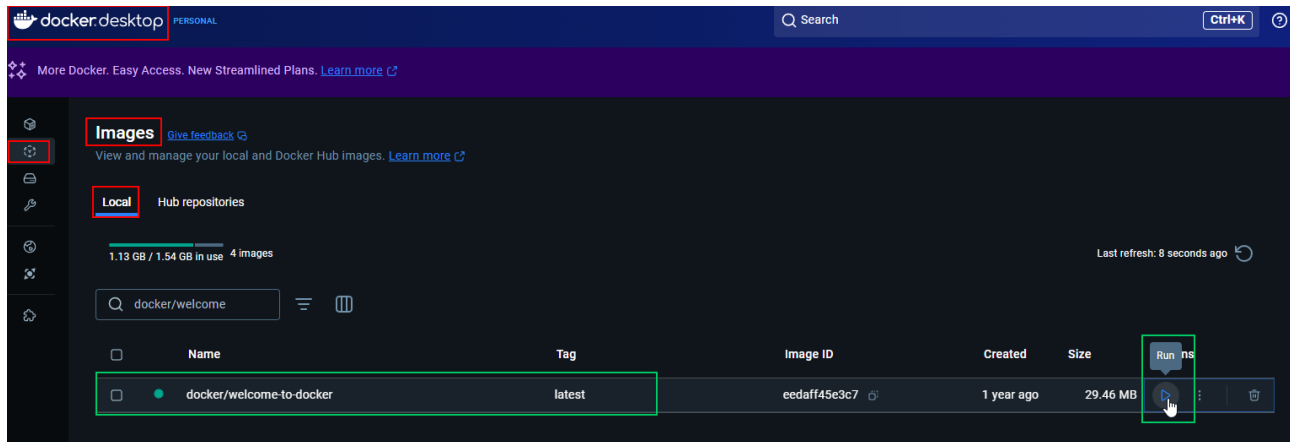
PS C:\Docker_jpls\welcome-to-docker> docker build -t welcome-to-docker .
[*] Building 3.0s (3/3)

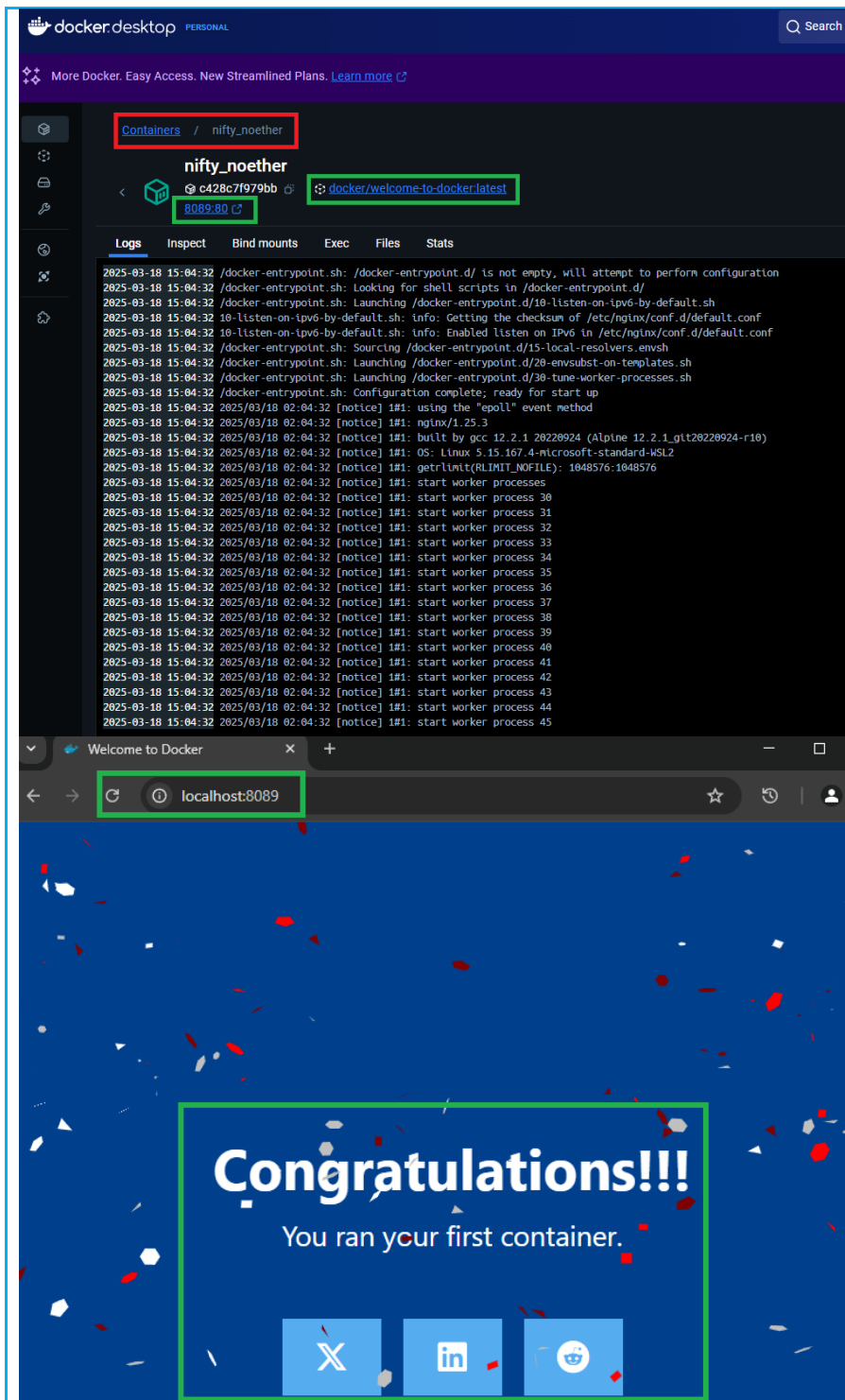
-> [internal] load build definition from Dockerfile
-> resolve docker.io/library/node:18-alpine@sha256:e0340f26173b41066d68e3fe9bfbdb6571ab3cad8a4272919a52e36f4ae56925 0.0s
-> resolve docker.io/library/node:18-alpine@sha256:e0340f26173b41066d68e3fe9bfbdb6571ab3cad8a4272919a52e36f4ae56925 0.0s
-> transferring dockerfile: 647B 0.0s
-> sha256:3d8c59f7308dbad7c144c50d7f2b78dbb5ee2f158009c4c17a8bdf56d74db0 444B / 444B 0.0s
-> [auth] library/node:pull token for registry-1.docker.io 0.0s
-> sha256:f46e519824fbb18a1df4becba40665f5c1cd193e527b3b0e26683d10f140916 1.20MB / 1.20MB 0.5s
-> [1/6] FROM docker.io/library/node:18-alpine@sha256:e0340f26173b41066d68e3fe9bfbdb6571ab3cad8a4272919a52e36f4ae56925 3.0s
-> sha256:a5cf150cb3746a8af43a656f15a671bb84fa704ac2413cf1e57cc37d12afa838 40.01MB / 40.01MB 1.1s
-> sha256:3d8c59f7308dbad7c144c50d7f2b78dbb5ee2f158009c4c17a8bdf56d74db0 444B / 444B 0.3s
-> extracting sha256:a5cf150cb3746a8af43a656f15a671bb84fa704ac2413cf1e57cc37d12afa838 0.8s
-> resolve docker.io/library/node:18-alpine@sha256:e0340f26173b41066d68e3fe9bfbdb6571ab3cad8a4272919a52e36f4ae56925 0.0s
-> extracting sha256:3d8c59f7308dbad7c144c50d7f2b78dbb5ee2f158009c4c17a8bdf56d74db0 444B / 444B 0.0s
-> sha256:3d8c59f7308dbad7c144c50d7f2b78dbb5ee2f158009c4c17a8bdf56d74db0 444B / 444B 0.0s
-> extracting sha256:f46e519824fbb18a1df4becba40665f5c1cd193e527b3b0e26683d10f140916 1.20MB / 1.20MB 0.1s
-> [4/6] COPY ./src ./src 0.1s
-> sha256:a5cf150cb3746a8af43a656f15a671bb84fa704ac2413cf1e57cc37d12afa838 40.01MB / 40.01MB 0.1s
-> transferring context: 691.39kB 0.0s
-> extracting sha256:a5cf150cb3746a8af43a656f15a671bb84fa704ac2413cf1e57cc37d12afa838 0.0s
-> resolve docker.io/library/node:18-alpine@sha256:e0340f26173b41066d68e3fe9bfbdb6571ab3cad8a4272919a52e36f4ae56925 0.0s
-> [5/6] COPY package*.json ./ 0.1s
-> sha256:3d8c59f7308dbad7c144c50d7f2b78dbb5ee2f158009c4c17a8bdf56d74db0 444B / 444B 0.0s
-> sha256:f46e519824fbb18a1df4becba40665f5c1cd193e527b3b0e26683d10f140916 1.20MB / 1.20MB 0.1s
-> [internal] load build context
-> transferring context: 52B 0.0s
[*] Building 52.5s (12/12) FINISHED

-> [internal] load build definition from Dockerfile
-> resolving manifest sha256:c36647e66904f7f246b3f85fb7aad7980bfdd158c40e5262b4646b6abc5f 0.0s
-> exporting config sha256:63979c85110897d40e4040a0e571c727d4d8a6f996cc3a0570f80c3706 0.0s
-> exporting attestation manifest sha256:704d99747f282f80d56a74bdeb27644332326e2d0970fc4818bfd215044 0.0s
-> exporting manifest list sha256:53c058ff61fa7a4f9203839c0e0dbcc1e4f996cd9f0219f13a99ced094293e 0.0s
-> naming to docker.io/library/welcome-to-docker:latest 0.0s
-> unpacking to docker.io/library/welcome-to-docker:latest 1.7s

```

- `Docker build -t welcome-to-docker .`





- Push Docker Image to Docker Hub.

```

PS C:\Docker_jpls\welcome-to-docker> docker images --filter=reference=welcome-to-docker
REPOSITORY          TAG         IMAGE ID      CREATED        SIZE
welcome-to-docker    latest      53c050ff61fa  23 minutes ago 383MB
PS C:\Docker_jpls\welcome-to-docker> docker tag welcome-to-docker dockerlsp123/welcome-to-docker-jpls:latest

```

- Docker images
- Docker tag welcome-to-docker dockerlsp123/welcome-to-docker-jpls:latest

```

PS C:\Docker_jpls\welcome-to-docker> docker images --filter=reference=dockerlsp123/welcome-to-docker-jpls
REPOSITORY          TAG         IMAGE ID      CREATED        SIZE
dockerlsp123/welcome-to-docker-jpls  latest      53c050ff61fa  24 minutes ago 383MB
PS C:\Docker_jpls\welcome-to-docker>

```

```

PS C:\Docker_jpls\welcome-to-docker> docker login
Authenticating with existing credentials...
Login Succeeded

```

```

PS C:\Docker_jpls\welcome-to-docker> docker push dockerlsp123/welcome-to-docker-jpls:latest
The push refers to repository [docker.io/dockerlsp123/welcome-to-docker-jpls]
f18232174bc9: Mounted from dockerlsp123/devops-jeanphilippels
f46e519824fb: Pushing [=====>] 1.261MB/1.261MB
f7f4b521746b: Pushed
31e2774c5eb0: Pushing [====>] 4.194MB/62.88MB
3d8c59f7308d: Pushed
a5cf150cb374: Pushing [=====>] 6.291MB/40.01MB
6e0bb1434e7f: Pushed
e6f3dfa2eb18: Pushed
e28bbca3bfc7: Pushed
f9ec8daafabc: Pushed

```

- Docker push dockerlsp123/welcome-to-docker-jpls:latest

```
PS C:\Docker_jpls\welcome-to-docker> docker push dockerlsp123/welcome-to-docker-jpls:latest
The push refers to repository [docker.io/dockerlsp123/welcome-to-docker-jpls]
f18232174bc9: Mounted from dockerlsp123/devops-jeanphilippels
f46e519824fb: Pushed
f7f4b521746b: Pushed
31e2774c5eb0: Pushed
3d8c59f7308d: Pushed
a5cf150cb374: Pushed
6e0bb1434e7f: Pushed
e6f3dfa2eb18: Pushed
e28bbca3bfc7: Pushed
f9ec8daafabc: Pushed
latest: digest: sha256:53c050ff61fa7a4f9203839c60e8cbcc1e4f996cd9f6219f33ad99ced694293e size: 856
PS C:\Docker_jpls\welcome-to-docker>
```

Repositories / welcome-to-docker-jpls / General

dockerlsp123/welcome-to-docker-jpls

Last pushed 2 minutes ago · Repository size: 103 MB

Add a description

Add a category

General Tags Image Management **BETA** Collaborators Webhooks Settings

Tags

This repository contains 1 tag(s).

Tag	OS	Type	Pulled	Pushed
latest		Image	less than 1 day	2 minutes

[See all](#)

- Building DevOps image with Dockerfile

```
1 FROM nginx:alpine
2
3 COPY ./src /usr/share/nginx/html
4
5 EXPOSE 80
```

```
PS C:\Docker_jpls> docker login
Authenticating with existing credentials...
Login did not succeed, error: error during connect: Post "http://%2f%2F.%2FdockerDesktopLinuxEngine/v1.47/auth": open //.pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

USING WEB-BASED LOGIN
To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: VPMH-NZHQ
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...
█
```



Device Confirmation

Please confirm this is the code displayed on your Docker CLI:

VMHW-NZHQ

If you did not initiate this action or you do not recognize this device select cancel.

Cancel

Confirm

```
PS C:\Docker_jpls> docker build -t devops-apps-jeanphilippels .
[*] Building 6.2s (8/8) FINISHED
-> [internal] load build definition from Dockerfile
-> transferring dockerfile: 103B
-> [internal] load metadata for docker.io/library/nginx:alpine
-> [auth] library/nginx:pull token for registry-1.docker.io
-> [internal] load .dockerignore
-> transferring context: 2B
-> [internal] load build context
-> transferring context: 547.46kB
-> [1/2] FROM docker.io/library/nginx:alpine@sha256:4ff102c5d78d25a6f0da662b3cf39eaf07f01eec0927fd21e219d0af8bc0591
-> resolve docker.io/library/nginx:alpine@sha256:4ff102c5d78d25a6f0da662b3cf39eaf07f01eec0927fd21e219d0af8bc0591
-> sha256:6d79c6084d434876ce0f83c675d20532f28e476283a29e7e63bc0bf134ed6 1.40kB / 1.40kB
-> sha256:0c7e4c092ab716c95e1a8840943549b4ac5e6c5d1acc74496528e1b03e1e67a 15.38kB / 15.38kB
-> sha256:ab3286a7346303a31b09a5189f63f1414c1de4e397088dc07edb322df1fe9 1.21kB / 1.21kB
-> sha256:8d27c072a58f81ecf2425172ac8e5b25010ff2d014f80de35090104e462568eb 402B / 402B
-> sha256:984583bcf083fa6900b5e7834795a9a57a9b4dfe7448d5350474f5d309625ece 955B / 955B
-> sha256:43f2ec460bdf5abae56d7e13b060b308f42cbf2103f6665c349f8698dac3962 627B / 627B
-> sha256:ccc35e35d420d0dd115290f1074afc6ad1c474b6f94897434e6befd33be781d2 1.79MB / 1.79MB
-> sha256:f18232174bc91741fd3da96d85011092101a032a93a388b79e99e69c2d5c870 3.64MB / 3.64MB
-> extracting sha256:f18232174bc91741fd3da96d85011092101a032a93a388b79e99e69c2d5c870
-> extracting sha256:ccc35e35d420d0dd115290f1074afc6ad1c474b6f94897434e6befd33be781d2
-> extracting sha256:43f2ec460bdf5abae56d7e13b060b308f42cbf2103f6665c349f8698dac3962
-> extracting sha256:984583bcf083fa6900b5e7834795a9a57a9b4dfe7448d5350474f5d309625ece
-> extracting sha256:8d27c072a58f81ecf2425172ac8e5b25010ff2d014f80de35090104e462568eb
-> extracting sha256:ab3286a7346303a31b09a5189f63f1414c1de4e397088dc07edb322df1fe9
-> extracting sha256:6d79c6084d434876ce0f83c675d20532f28e476283a29e7e63bc0bf134ed6
-> extracting sha256:0c7e4c092ab716c95e1a8840943549b4ac5e6c5d1acc74496528e1b03e1e67a
-> [2/2] COPY ./src /usr/share/nginx/html
-> exporting to image
-> exporting layers
-> exporting manifest sha256:ea324e542456625fc075c95f664b929071b8d1f3e5cb33216a13a3e2dcfc41d4
-> exporting config sha256:716dc1bc10521e745bac7af0c0a800b0c0fda12c78c1cebd67beb1ddd79a8a65
-> exporting attestation manifest sha256:e3d0aded6e9310d0e40f486f76ba36b514c87ae4f346c9104c3e489305eb658
-> exporting manifest list sha256:1b5acc1844ee7b2bd4cac941f542828e4019cc833c84ee59ac9326f351b09e2d
-> naming to docker.io/library/devops-apps-jeanphilippels:latest
-> unpacking to docker.io/library/devops-apps-jeanphilippels:latest
```

- `docker build -t devops-apps-jeanphilippels .`

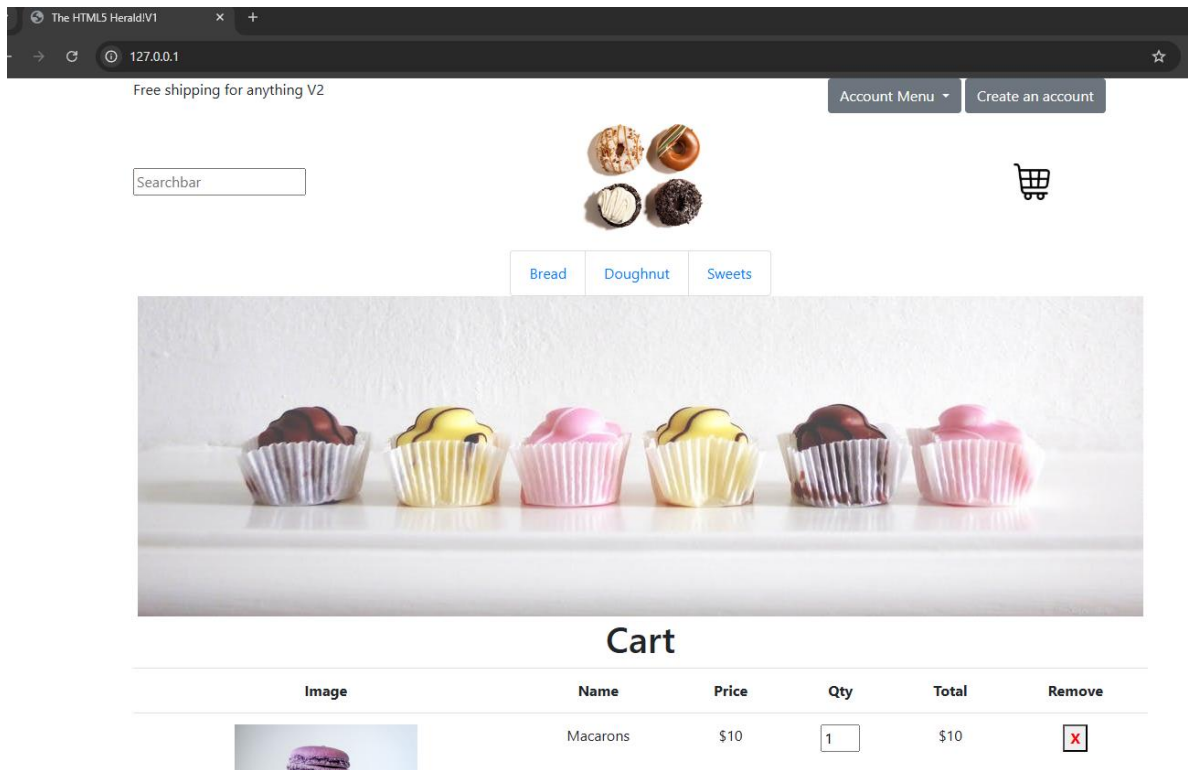
```
PS C:\Docker_jpls> docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
devops-apps-jeanphilippels latest          1b5acc1844ee    18 seconds ago 74MB
kindest/node         <none>         f226345927d7    3 weeks ago    1.49GB
PS C:\Docker_jpls>
```

- `docker images`
- Building container with the DevOps image

```
PS C:\Docker_jpls> docker run -d -p 80:80 --name devops-jeanphilippels-container devops-apps-jeanphilippels:latest
381164ffdc150bc0fed607615b8172fb8af38a20c03084e32c9336b452399d0c
PS C:\Docker_jpls> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                    NAMES
381164ffdc15   devops-apps-jeanphilippels:latest   "/docker-entrypoint..." 5 seconds ago  Up 5 seconds  0.0.0.0:80->80/tcp      devops-jeanphilippels-container
PS C:\Docker_jpls>
```

- `docker run -d -p 80:80 --name devops-jeanphilippels-container devops-apps-jeanphilippels:latest`
- `docker ps`

- Testing container with the DevOps image



- Get into the DevOps docker container

```
PS C:\Docker_jpls> docker exec -it devops-jeanphilippels-container sh
/ # ls -la
total 80
drwxr-xr-x 1 root root 4096 Mar 14 03:33 .
drwxr-xr-x 1 root root 4096 Mar 14 03:33 ..
-rwxr-xr-x 1 root root 0 Mar 14 03:33 .dockerenv
drwxr-xr-x 2 root root 4096 Feb 13 23:04 bin
drwxr-xr-x 5 root root 340 Mar 14 03:33 dev
drwxr-xr-x 1 root root 4096 Feb 14 19:16 docker-entrypoint.d
-rwxr-xr-x 1 root root 1620 Feb 14 19:16 docker-entrypoint.sh
drwxr-xr-x 1 root root 4096 Mar 14 03:33 etc
drwxr-xr-x 2 root root 4096 Feb 13 23:04 home
drwxr-xr-x 1 root root 4096 Feb 13 23:04 lib
drwxr-xr-x 5 root root 4096 Feb 13 23:04 media
drwxr-xr-x 2 root root 4096 Feb 13 23:04 mnt
drwxr-xr-x 2 root root 4096 Feb 13 23:04 opt
dr-xr-xr-x 298 root root 0 Mar 14 03:33 proc
drwx----- 1 root root 4096 Mar 14 03:36 root
drwxr-xr-x 1 root root 4096 Mar 14 03:33 run
drwxr-xr-x 2 root root 4096 Feb 13 23:04/sbin
drwxr-xr-x 2 root root 4096 Feb 13 23:04/srv
dr-xr-xr-x 11 root root 0 Mar 14 03:33/sys
drwxrwxrwt 2 root root 4096 Feb 13 23:04/tmp
drwxr-xr-x 1 root root 4096 Feb 13 23:04/usr
drwxr-xr-x 1 root root 4096 Feb 13 23:04/var
```


- `docker exec -it devops-apps-jeanphilippels:latest sh`

- Pushing DevOps image to private docker repository

```
PS C:\Docker_jpls> docker images
REPOSITORY          TAG          IMAGE ID      CREATED        SIZE
devops-apps-jeanphilippels  latest      1b5acc1844ee  About a minute ago  74MB
kindest/node         <none>      f226345927d7  3 weeks ago    1.49GB
PS C:\Docker_jpls> docker tag devops-apps-jeanphilippels dockerlspf123/devops-jeanphilippels:manual1
PS C:\Docker_jpls> docker images
REPOSITORY          TAG          IMAGE ID      CREATED        SIZE
dockerlspf123/devops-jeanphilippels  manual1      1b5acc1844ee  3 minutes ago  74MB
devops-apps-jeanphilippels          latest      1b5acc1844ee  3 minutes ago  74MB
kindest/node                        <none>      f226345927d7  3 weeks ago    1.49GB
PS C:\Docker_jpls> []
```

- `docker images`

```
PS C:\Docker_jpls> docker images
REPOSITORY          TAG          IMAGE ID      CREATED        SIZE
devops-apps-jeanphilippels  latest      1b5acc1844ee  About a minute ago  74MB
kindest/node         <none>      f226345927d7  3 weeks ago    1.49GB
PS C:\Docker_jpls> docker tag devops-apps-jeanphilippels dockerlspf123/devops-jeanphilippels:manual1
PS C:\Docker_jpls> docker images
REPOSITORY          TAG          IMAGE ID      CREATED        SIZE
dockerlspf123/devops-jeanphilippels  manual1      1b5acc1844ee  3 minutes ago  74MB
devops-apps-jeanphilippels          latest      1b5acc1844ee  3 minutes ago  74MB
kindest/node                        <none>      f226345927d7  3 weeks ago    1.49GB
PS C:\Docker_jpls> []
```

- `docker tags devops-apps-jeanphilippels dockerlspf123/devops-jeanphilippels:manual1`
- `Docker images`

```

PS C:\Docker_jpls> docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
devops-apps-jeanphilippels  latest         1b5acc1844ee   About a minute ago  74MB
kindest/node         <none>         f226345927d7   3 weeks ago     1.49GB
PS C:\Docker_jpls> docker tag devops-apps-jeanphilippels dockerlsp123/devops-jeanphilippels:manual1
PS C:\Docker_jpls> docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
dockerlsp123/devops-jeanphilippels  manual1         1b5acc1844ee   3 minutes ago   74MB
devops-apps-jeanphilippels         latest         1b5acc1844ee   3 minutes ago   74MB
kindest/node                       <none>         f226345927d7   3 weeks ago     1.49GB
PS C:\Docker_jpls> docker push dockerlsp123/devops-jeanphilippels:manual1
The push refers to repository [docker.io/dockerlsp123/devops-jeanphilippels]
ab3286a73463: Layer already exists
8d27c072a58f: Layer already exists
f18232174bc9: Layer already exists
43f2ec460bdf: Layer already exists
6d79cc6084d4: Layer already exists
98bcf5d5f13a: Pushed
01637544f41d: Pushed
984583bcf083: Layer already exists
ccc35e35d420: Layer already exists
0c7e4c092ab7: Layer already exists
manual1: digest: sha256:1b5acc1844ee7b2bd4cac941f542828e4019cc033c04ee59ac9326f351b89e2d size: 856
PS C:\Docker_jpls>

```

- `docker push dockerlsp123/devops-jeanphilippels:manual1`

The screenshot shows the Docker Hub interface for the repository `dockerlsp123/devops-jeanphilippels`. The page includes a navigation bar with 'Explore', 'Repositories', 'Organizations', and 'Usage'. The repository page shows the name, a lock icon, and the last push time. Below the repository name, there are tabs for 'General', 'Tags', 'Image Management', 'Collaborators', 'Webhooks', and 'Settings'. The 'Tags' tab is selected, showing a table of tags. The table has columns for 'TAG', 'Last pushed 1 minute by', 'Digest', 'OS/ARCH', 'Last pull', and 'Compressed size'. The 'manual1' tag is listed with a digest of `ea324e542456` and a compressed size of 20.17 MB. A 'docker pull' command is shown with a 'Copy' button.

Deployment in Local Kubernetes Clusters

- **Deploy the container using Kubernetes / Apply Kubernetes Configuration**
- *Configure cluster*
- *Create a namespace*
- *Create a nginx deployment*
- *Create a service nginx nodeport*
- *Create a scale of the nginx deployment*

- Deploy the container using Kubernetes / Apply Kubernetes Configuration
- *Create the custom namespace for devops applications*
 - *Create the Secrets allowing to use private docker repository*
 - *Create the service account*
 - *Create the role cluster for the service account*
 - *Create the pod's role for the service account*
- *Bind the role cluster and pod's role to the service account*
- *Create the custom devops web application deployment with secret from a docker hub repository*
- *Create the service custom devops web application deployment*
- *Testing the devops web deployment*

Kind (Kubernetes IN Docker) is a tool allowing to setup a local Kubernetes clusters using Docker Containers. Local development can be explored by using this tool.

Kubernetes is a container orchestration platform allows to automates deployment, the scaling, and the management of containerized applications

- **Configure cluster**

```
PS C:\kubernetes> kind create cluster --config .\kind.yaml --name cka-cluster-jeanphilippels
Creating cluster "cka-cluster-jeanphilippels" ...
  • Ensuring node image (kindest/node:v1.32.2) ...
  ✓ Ensuring node image (kindest/node:v1.32.2) ...
  • Preparing nodes ...
  ✓ Preparing nodes ...
  • Writing configuration ...
  ✓ Writing configuration ...
  • Starting control-plane ...
  ✓ Starting control-plane ...
  • Installing CNI ...
  ✓ Installing CNI ...
  • Installing StorageClass ...
  ✓ Installing StorageClass ...
  • Joining worker nodes ...
  ✓ Joining worker nodes ...

Set kubectl context to "kind-cka-cluster-jeanphilippels"
You can now use your cluster with:

kubectl cluster-info --context kind-cka-cluster-jeanphilippels

Have a nice day! 🍀
```

```
! kind.yaml
1 # three node (two workers) cluster config
2 kind: Cluster
3 apiVersion: kind.x-k8s.io/v1alpha4
4 nodes:
5 - role: control-plane
6   extraPortMappings:
7     - containerPort: 30001
8       hostPort: 30001
9 - role: worker
10 - role: worker
```

```
Kind create cluster -config .\kind.yaml -name cka-cluster-jeanphilippels
```

```
PS C:\kubernetes> kind get clusters
cka-cluster-jeanphilippels
PS C:\kubernetes>
```

- **Kubectl get clusters**

```
PS C:\kubernetes> kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
cka-cluster-jeanphilippels-control-plane	Ready	control-plane	58m	v1.32.2
cka-cluster-jeanphilippels-worker	Ready	<none>	58m	v1.32.2
cka-cluster-jeanphilippels-worker2	Ready	<none>	58m	v1.32.2

```
PS C:\kubernetes>
```

- **Create a namespace**

Kubernetes namespaces allow to organize partition and organize resources within a Kubernetes cluster. They separate in logical groups of resources within a cluster, making cluster management more efficient. They can denote multiple environments like dev, staging, or production, or for grouping resources that make up an application.

- **Create a nginx deployment**

```
PS C:\kubernetes> kubectl create namespace nginx-apps-jeanphilippels
namespace/nginx-apps-jeanphilippels created
PS C:\kubernetes> kubectl apply -f .\nginx-deploy.yml
deployment.apps/nginx-deploy-jeanphilippels created
PS C:\kubernetes> kubectl apply -f .\nginx-nodeport.yml
service/nginx-nodeport-jeanphilippels created
PS C:\kubernetes> kubectl get deploy -n nginx-apps-jeanphilippels
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
nginx-deploy-jeanphilippels	3/3	3	3	23s

```
PS C:\kubernetes>
```

```
PS C:\kubernetes> kubectl get namespace -n nginx-apps-jeanphilippels
```

NAME	STATUS	AGE
default	Active	5h26m
dev-apps-jeanphilippels	Active	5h20m
kube-node-lease	Active	5h26m
kube-public	Active	5h26m
kube-system	Active	5h26m
local-path-storage	Active	5h26m
nginx-apps-jeanphilippels	Active	4h8m

```
PS C:\kubernetes> |
```

- Kubectl create namespace nginx-apps-jeanphilippels
- Kubectl apply -f .\nginx-deploy.yml
- Kubectl apply -f .\nginx-nodeport.yml
- Kubectl get deploy -n nginx-apps-jeanphilippels

```
PS C:\kubernetes> kubectl get pods -n nginx-apps-jeanphilippels
```

NAME	READY	STATUS	RESTARTS	AGE
nginx-deploy-jeanphilippels-5f6bd4879d-4cftr	1/1	Running	0	50s
nginx-deploy-jeanphilippels-5f6bd4879d-gxwqn	1/1	Running	0	50s
nginx-deploy-jeanphilippels-5f6bd4879d-mwvss	1/1	Running	0	50s

```
PS C:\kubernetes> |
```

- Kubectl get pods -n nginx-apps-jeanphilippels

```
PS C:\kubernetes> kubectl edit deploy/nginx-deploy-jeanphilippels -n nginx-apps-jeanphilippels
```

- Example of increase replicas with “edit”

- Kubectl edit deploy/nginx-deploy-jeanphilippels -n nginx-apps-jeanphilippels

The image displays two screenshots of a Notepad window editing a Kubernetes Deployment YAML file. The file name is 'kubectl.exe-edit-3503174900.yaml'. The content is a YAML configuration for a Deployment.

First Screenshot (Initial State):

```
# Please edit the object below. Lines beginning with a '#' will be ignored,
# and an empty file will abort the edit. If an error occurs while saving this file w
# reopened with the relevant failures.
#
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations:
    deployment.kubernetes.io/revision: "1"
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"apps/v1","kind":"Deployment","metadata":{"annotations":{},"labe
creationTimestamp: "2025-03-10T22:16:49Z"
generation: 1
labels:
  env: demo
  name: nginx-deploy-jeanphilippels
  namespace: nginx-apps-jeanphilippels
  resourceVersion: "8152"
  uid: 842eea47-ea30-4bcc-8ca0-db536af5f06f
spec:
  progressDeadlineSeconds: 600
  replicas: 3
  revisionHistoryLimit: 10
  selector:
    matchLabels:
      env: demo
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
```

Second Screenshot (Edited State):

```
# Please edit the object below. Lines beginning with a '#' will be ignored,
# and an empty file will abort the edit. If an error occurs while saving this file w
# reopened with the relevant failures.
#
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations:
    deployment.kubernetes.io/revision: "1"
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"apps/v1","kind":"Deployment","metadata":{"annotations":{},"labe
creationTimestamp: "2025-03-10T22:16:49Z"
generation: 1
labels:
  env: demo
  name: nginx-deploy-jeanphilippels
  namespace: nginx-apps-jeanphilippels
  resourceVersion: "8152"
  uid: 842eea47-ea30-4bcc-8ca0-db536af5f06f
spec:
  progressDeadlineSeconds: 600
  replicas: 5
  revisionHistoryLimit: 10
  selector:
    matchLabels:
      env: demo
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
```

```
PS C:\kubernetes> kubectl get deploy -n nginx-apps-jeanphilippels
NAME                                READY  UP-TO-DATE  AVAILABLE  AGE
nginx-deploy-jeanphilippels        5/5    5            5           4m40s
PS C:\kubernetes>
```

- **Kubectl get deploy -n nginx-apps-jeanphilippels**

```
PS C:\kubernetes> kubectl describe deploy/nginx-deploy-jeanphilippels -n nginx-apps-jeanphilippels
Name:                               nginx-deploy-jeanphilippels
Namespace:                           nginx-apps-jeanphilippels
CreationTimestamp:                   Tue, 11 Mar 2025 11:16:49 +1300
Labels:                               env=demo
Annotations:                         deployment.kubernetes.io/revision: 1
Selector:                             env=demo
Replicas:                            5 desired | 5 updated | 5 total | 5 available | 0 unavailable
StrategyType:                        RollingUpdate
MinReadySeconds:                     0
RollingUpdateStrategy:               25% max unavailable, 25% max surge
Pod Template:
  Labels:                             env=demo
  Containers:
    nginx:
      Image:                           nginx
      Port:                             <none>
      Host Port:                       <none>
      Environment:                     <none>
      Mounts:                           <none>
      Volumes:                           <none>
      Node-Selectors:                   <none>
      Tolerations:                       <none>
Conditions:
  Type           Status  Reason
  ----           -
  Progressing    True    NewReplicaSetAvailable
  Available      True    MinimumReplicasAvailable
OldReplicaSets: <none>
NewReplicaSet:  nginx-deploy-jeanphilippels-5f6bd4879d (5/5 replicas created)
Events:
  Type           Reason              Age             From              Message
  ----           -
  Normal         ScalingReplicaSet   6m3s            deployment-controller Scaled up replica set nginx-deploy-jeanphilippels-5f6bd4879d from 0 to 3
  Normal         ScalingReplicaSet   92s             deployment-controller Scaled up replica set nginx-deploy-jeanphilippels-5f6bd4879d from 3 to 5
PS C:\kubernetes>
```

- **Kubectl describe deploy/nginx-deploy-jeanphilippels**

- **Example of increase replicas with “scale”**

```
PS C:\kubernetes> kubectl get deploy -n nginx-apps-jeanphilippels
NAME                                READY  UP-TO-DATE  AVAILABLE  AGE
nginx-deploy-jeanphilippels        5/5    5            5           3h33m
PS C:\kubernetes> kubectl scale deploy/nginx-deploy-jeanphilippels -n nginx-apps-jeanphilippels --replicas=10
deployment.apps/nginx-deploy-jeanphilippels scaled
PS C:\kubernetes> kubectl get deploy -n nginx-apps-jeanphilippels
NAME                                READY  UP-TO-DATE  AVAILABLE  AGE
nginx-deploy-jeanphilippels        7/10   10           7           3h34m
PS C:\kubernetes> kubectl get deploy -n nginx-apps-jeanphilippels
NAME                                READY  UP-TO-DATE  AVAILABLE  AGE
nginx-deploy-jeanphilippels        10/10  10           10          3h34m
```

- **Kubectl get deploy -n nginx-apps-jeanphilippels**
- **Kubectl scale deploy/nginx-deploy-jeanphilippels -n nginx-apps-jeanphilippels --replicas=10**

- **Kubectl get deploy -n nginx-apps-jeanphilippels**

```
PS C:\kubernetes> kubectl get pods -n nginx-apps-jeanphilippels
NAME                                     READY   STATUS    RESTARTS   AGE
nginx-deploy-jeanphilippels-5f6bd4879d-4cftr  1/1     Running   1 (99s ago)  3h35m
nginx-deploy-jeanphilippels-5f6bd4879d-4znh4  1/1     Running   0           58s
nginx-deploy-jeanphilippels-5f6bd4879d-7rv42  1/1     Running   0           58s
nginx-deploy-jeanphilippels-5f6bd4879d-g6x8l  1/1     Running   0           58s
nginx-deploy-jeanphilippels-5f6bd4879d-gxwqn  1/1     Running   1 (99s ago)  3h35m
nginx-deploy-jeanphilippels-5f6bd4879d-lqldx  1/1     Running   1 (99s ago)  3h30m
nginx-deploy-jeanphilippels-5f6bd4879d-mwvss  1/1     Running   1 (99s ago)  3h35m
nginx-deploy-jeanphilippels-5f6bd4879d-sndnz  1/1     Running   1 (99s ago)  3h30m
nginx-deploy-jeanphilippels-5f6bd4879d-t6rzt  1/1     Running   0           58s
nginx-deploy-jeanphilippels-5f6bd4879d-vt8cz  1/1     Running   0           58s
PS C:\kubernetes>
```

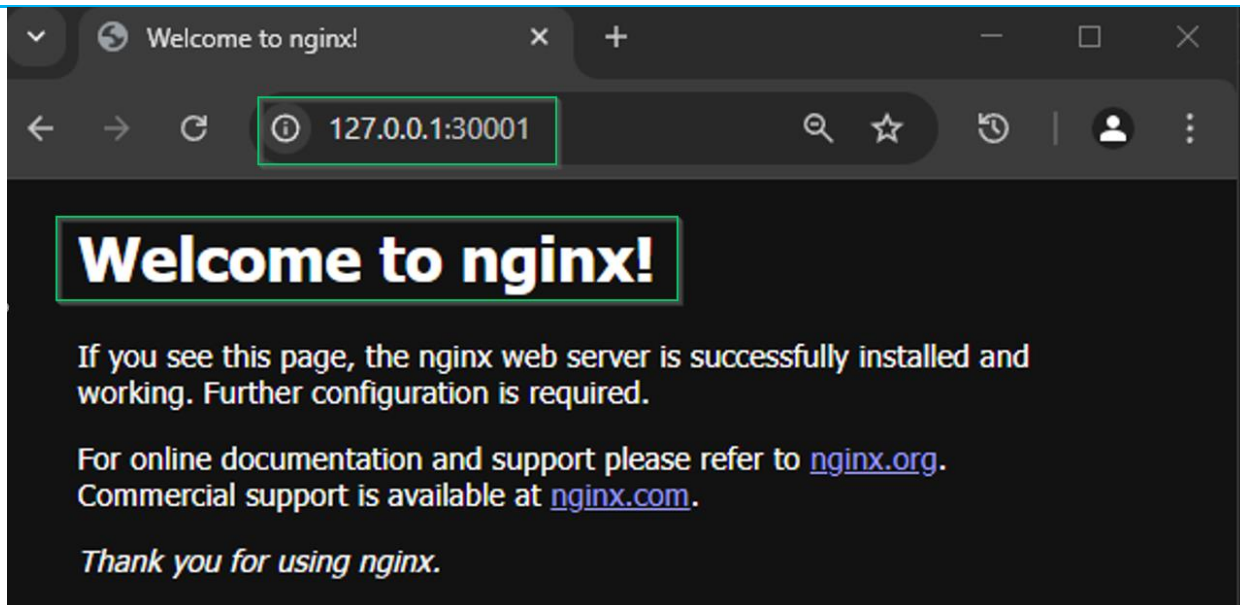
- **Kubectl get pods -n nginx-apps-jeanphilippels**

```
PS C:\kubernetes> kubectl get svc -n nginx-apps-jeanphilippels
NAME                                TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)          AGE
nginx-nodeport-jeanphilippels      NodePort    10.96.81.182  <none>       80:30001/TCP    4h5m
PS C:\kubernetes>
```

- **Delete a nginx deployment**

```
PS C:\kubernetes> kubectl delete deploy/nginx-deploy-jeanphilippels -n nginx-apps-jeanphilippels
deployment.apps "nginx-deploy-jeanphilippels" deleted
PS C:\kubernetes> kubectl get pods -n nginx-apps-jeanphilippels
No resources found in nginx-apps-jeanphilippels namespace.
PS C:\kubernetes>
```

- **Kubectl delete deploy/nginx-deploy-jeanphilippes -n nginx-apps-jeanphilippels**



- Create the custom namespace for devops apps

```
PS C:\kubernetes> kubectl create namespace dev-apps-jeanphilippels
namespace/dev-apps-jeanphilippels created
PS C:\kubernetes> kubectl get namespace
NAME                STATUS  AGE
default             Active  6m57s
dev-apps-jeanphilippels  Active  9s
kube-node-lease     Active  6m57s
kube-public         Active  6m57s
kube-system         Active  6m58s
local-path-storage  Active  6m49s
PS C:\kubernetes>
```

- Create the Secrets allowing to use private docker repository

```
PS C:\kubernetes> kubectl get secrets
No resources found in default namespace.
PS C:\kubernetes> kubectl get secrets -n dev-apps-jeanphilippels
No resources found in dev-apps-jeanphilippels namespace.
PS C:\kubernetes> kubectl create secret docker-registry devops-dev-sec-jeanphilippels --docker-server=docker.io --docker-username=dockerlsp123 --docker-password=jeanphilippels@hotmail.com --docker-email=jeanphilippels@hotmail.com -n dev-apps-jeanphilippels
secret/devops-dev-sec-jeanphilippels created
PS C:\kubernetes> kubectl get secrets -n dev-apps-jeanphilippels
NAME                TYPE              DATA  AGE
devops-dev-sec-jeanphilippels  kubernetes.io/dockerconfigjson  1      4s
PS C:\kubernetes>
```

- `kubectl create secret docker-registry devops-dev-sec-jeanphilippels --docker-server=docker.io --docker-username=dockerlsp123 --docker-password=<PATAACCESTOKEN> --docker-email=jeanphilippels@hotmail.com -n dev-apps-jeanphilippels`

- Create the service account

```
PS C:\kubernetes> kubectl get sa -n dev-apps-jeanphilippels
NAME      SECRETS  AGE
default   0         6m46s
PS C:\kubernetes> kubectl get sa
NAME      SECRETS  AGE
default   0         13m
PS C:\kubernetes> kubectl apply -f .\serviceaccount.yml
serviceaccount/sa-jpls created
PS C:\kubernetes> kubectl get sa -n dev-apps-jeanphilippels
NAME      SECRETS  AGE
default   0         6m57s
sa-jpls    0         2s
PS C:\kubernetes>
```

- `kubectl apply -f .\serviceaccount.yml -n dev-apps-jeanphilippels`

- Create the role cluster for the service account
- Create the pod's role for the service account
- Bind the role cluster and pod's role to the service account

```
PS C:\kubernetes> kubectl get role -n dev-apps-jeanphilippels
No resources found in dev-apps-jeanphilippels namespace.
PS C:\kubernetes> kubectl get rolebinding -n dev-apps-jeanphilippels
No resources found in dev-apps-jeanphilippels namespace.
PS C:\kubernetes> kubectl apply -f .\role_cluster.yml
clusterrole.rbac.authorization.k8s.io/secret-reader-cluster-jeanphilippels created
PS C:\kubernetes> kubectl apply -f .\role_pod.yml
role.rbac.authorization.k8s.io/pod-reader-jeanphilippels created
PS C:\kubernetes> kubectl apply -f .\rolebinding-cluster.yml
rolebinding.rbac.authorization.k8s.io/read-secrets-jeanphilippels created
PS C:\kubernetes> kubectl apply -f .\rolebinding-sa.yml
rolebinding.rbac.authorization.k8s.io/read-pods-jeanphilippels created
PS C:\kubernetes> kubectl get role -n dev-apps-jeanphilippels
NAME                                CREATED AT
pod-reader-jeanphilippels           2025-03-10T21:36:33Z
PS C:\kubernetes> kubectl get rolebinding -n dev-apps-jeanphilippels
NAME                                ROLE                                AGE
read-pods-jeanphilippels            Role/pod-reader-jeanphilippels      23s
read-secrets-jeanphilippels         Role/secret-reader-jeanphilippels   31s
PS C:\kubernetes> kubectl describe clusterrole secret-reader-cluster-jeanphilippels
Name:                secret-reader-cluster-jeanphilippels
Labels:               <none>
Annotations:          <none>
PolicyRule:
  Resources            Non-Resource URLs  Resource Names      Verbs
  -----
  secrets              []                 []                  [get watch list]
PS C:\kubernetes>
```

- `kubectl apply -f .\role_cluster.yml`

- `kubectl apply -f .\role_pod.yml`
- `kubectl apply -f .\rolebinding-cluster.yml`
- `kubectl apply -f .\rolebinding-sa.yml`
- `kubectl get role -n dev-apps-jeanphilippels`
- `kubectl get rolebinding -n dev-apps-jeanphilippels`
- `kubectl describe clusterrole secret-reader-cluster-jeanphilippels`

- **Create the custom devops web application deployment with secret from a docker hub repository**

```
PS C:\kubernetes> kubectl get pods -n dev-apps-jeanphilippels
No resources found in dev-apps-jeanphilippels namespace.
PS C:\kubernetes> kubectl apply -f .\deploy_secrets.yml
deployment.apps/devops-deploy-jeanphilippels created
PS C:\kubernetes> kubectl get pods -n dev-apps-jeanphilippels
```

NAME	READY	STATUS	RESTARTS	AGE
devops-deploy-jeanphilippels-644797c7c-j78jt	0/1	ContainerCreating	0	2s
devops-deploy-jeanphilippels-644797c7c-rfphk	0/1	ContainerCreating	0	2s
devops-deploy-jeanphilippels-644797c7c-wh5tn	0/1	ContainerCreating	0	2s

```
PS C:\kubernetes> kubectl get pods -n dev-apps-jeanphilippels
```

NAME	READY	STATUS	RESTARTS	AGE
devops-deploy-jeanphilippels-644797c7c-j78jt	1/1	Running	0	14s
devops-deploy-jeanphilippels-644797c7c-rfphk	1/1	Running	0	14s
devops-deploy-jeanphilippels-644797c7c-wh5tn	1/1	Running	0	14s

```
PS C:\kubernetes>
```

```
PS C:\kubernetes> kubectl get deploy/devops-deploy-jeanphilippels -n dev-apps-jeanphilippels
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
devops-deploy-jeanphilippels	3/3	3	3	2m1s

```
PS C:\kubernetes>
```

- `kubectl apply -f .\deploy_secrets.yml`
- `kubectl get pods -n dev-apps-jeanphilippels`
- `kubectl get deploy/devops-deploy-jeanphilippels -n dev-apps-jeanphilippels`

IMPLEMENTING INFRASTRUCTURE AS CODE (IAC)

Terraform

- **Configure terraform local backend: terraform init / terraform plan / terraform apply**
- **Destroy resources provisioned with terraform destroy**
- **Configure terraform backend with Microsoft Azure**

Terraform is an Infrastructure as Code tool that allow to provision automatically infrastructures in the cloud and on-premises. It manages resources and configurations across multiple cloud environments.

- **Configure terraform local backend: terraform init / terraform plan / terraform apply**

The screenshot shows a VS Code editor with a file explorer on the left listing Terraform files: .terraform, .terraform.lock.hcl, locals.tf, main.tf, outputs.tf, provider.tf, variables.tf, and versions.tf. The main editor displays the content of main.tf, which defines an Azure Resource Group and an Azure Storage Account. The terminal at the bottom shows the output of the 'terraform init' command, including the initialization of the local backend and provider plugins.

```
main.tf > resource "azurerm_storage_account" "mrgpjeanphilipps" { location
1 resource "azurerm_resource_group" "mrgpjeanphilipps" {
2   name = var.rg-name
3   location = var.location
4 }
5
6 resource "azurerm_storage_account" "mrgpjeanphilipps" {
7
8   name = var.sact-name
9   resource_group_name = azurerm_resource_group.mrgpjeanphilipps.name
10  location = azurerm_resource_group.mrgpjeanphilipps.location
11  account_tier = var.sact-tier
12  account_replication_type = var.sact-rep-type
13  tags = {
14    environment = local.common_tags.environment
15    owner = local.common_tags.Owner
16  }
17 }
18
```

```
PS C:\terraform\ResourceGroup> terraform init
Initializing the backend...
Initializing provider plugins...
- Finding hashicorp/azurerm versions matching "~> 4.8.0"...
- Installing hashicorp/azurerm v4.8.0...
- Installed hashicorp/azurerm v4.8.0 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\terraform\ResourceGroup>
```

The screenshot shows the terminal output of the 'terraform plan' command. It indicates that Terraform will create the 'azurerm_resource_group.mrgpjeanphilipps' resource. The output is summarized in a table below.

Resource	Actions
azurerm_resource_group.mrgpjeanphilipps	create

```
PS C:\terraform\ResourceGroup> terraform plan --out tfplanjeanphilipps

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# azurerm_resource_group.mrgpjeanphilipps will be created
+ resource "azurerm_resource_group" "mrgpjeanphilipps" {
+   id = (known after apply)
+   location = "eastus"
+   name = "rg-tf-statefiles-jeanphilipps"
}
```

```

# azure_rm_storage_account.mrgpjeanphilippels will be created
+ resource "azure_rm_storage_account" "mrgpjeanphilippels" {
  + access_tier                        = (known after apply)
  + account_kind                      = "StorageV2"
  + account_replication_type          = "LRS"
  + account_tier                      = "Standard"
  + allow_nested_items_to_be_public   = true
  + cross_tenant_replication_enabled   = false
  + default_to_oauth_authentication    = false
  + dns_endpoint_type                 = "Standard"
  + https_traffic_only_enabled         = true
  + id                                = (known after apply)
  + infrastructure_encryption_enabled = false
  + is_hns_enabled                    = false
  + large_file_share_enabled           = (known after apply)
  + local_user_enabled                 = true
  + location                          = "eastus"
  + min_tls_version                   = "TLS1_2"
  + name                              = "tfstatesacntjpls"
  + nfsv3_enabled                     = false
  + primary_access_key                 = (sensitive value)
  + primary_blob_connection_string     = (sensitive value)
  + primary_blob_endpoint              = (known after apply)
  + primary_blob_host                  = (known after apply)
  + primary_blob_internet_endpoint     = (known after apply)
  + primary_blob_internet_host         = (known after apply)
  + primary_blob_microsoft_endpoint    = (known after apply)
  + primary_blob_microsoft_host        = (known after apply)
  + primary_connection_string          = (sensitive value)
  + primary_dfs_endpoint               = (known after apply)
  + primary_dfs_host                   = (known after apply)
  + primary_dfs_internet_endpoint      = (known after apply)
  + primary_dfs_internet_host          = (known after apply)
  + primary_dfs_microsoft_endpoint     = (known after apply)
  + primary_dfs_microsoft_host         = (known after apply)
  + primary_file_endpoint              = (known after apply)
  + primary_file_host                  = (known after apply)
  + primary_file_internet_endpoint     = (known after apply)
  + primary_file_internet_host         = (known after apply)
  + primary_file_microsoft_endpoint    = (known after apply)
  + primary_file_microsoft_host        = (known after apply)
  + primary_location                   = (known after apply)
  + primary_queue_endpoint             = (known after apply)
  + primary_queue_host                 = (known after apply)
  + primary_queue_microsoft_endpoint   = (known after apply)
  + primary_queue_microsoft_host       = (known after apply)
  + primary_table_endpoint            = (known after apply)
  + primary_table_host                 = (known after apply)
  + primary_table_microsoft_endpoint   = (known after apply)

```

```

+ secondary_file_host = (known after apply)
+ secondary_file_internet_endpoint = (known after apply)
+ secondary_file_internet_host = (known after apply)
+ secondary_file_microsoft_endpoint = (known after apply)
+ secondary_file_microsoft_host = (known after apply)
+ secondary_location = (known after apply)
+ secondary_queue_endpoint = (known after apply)
+ secondary_queue_host = (known after apply)
+ secondary_queue_microsoft_endpoint = (known after apply)
+ secondary_queue_microsoft_host = (known after apply)
+ secondary_table_endpoint = (known after apply)
+ secondary_table_host = (known after apply)
+ secondary_table_microsoft_endpoint = (known after apply)
+ secondary_table_microsoft_host = (known after apply)
+ secondary_web_endpoint = (known after apply)
+ secondary_web_host = (known after apply)
+ secondary_web_internet_endpoint = (known after apply)
+ secondary_web_internet_host = (known after apply)
+ secondary_web_microsoft_endpoint = (known after apply)
+ secondary_web_microsoft_host = (known after apply)
+ sftp_enabled = false
+ shared_access_key_enabled = true
+ table_encryption_key_type = "Service"
+ tags = {
  + "environment" = "dev"
  + "owner" = "IT"
}

+ blob_properties (known after apply)

+ network_rules (known after apply)

+ queue_properties (known after apply)

+ routing (known after apply)

+ share_properties (known after apply)
}

```

Plan: 2 to add, 0 to change, 0 to destroy.

Changes to Outputs:

+ storage_account_name = "tfstatesacctjpls"

Saved the plan to: tfplanjeanphilippls

To perform exactly these actions, run the following command to apply:
 terraform apply "tfplanjeanphilippls"

```
PS C:\terraform\ResourceGroup> terraform apply -tfplan=jeanphilipps
azurerm_resource_group.mrgpjeanphilipps: Creating...
azurerm_resource_group.mrgpjeanphilipps: Still creating... [10s elapsed]
azurerm_resource_group.mrgpjeanphilipps: Creation complete after 13s [id=/subscriptions/8de89c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-tf-statefiles-jeanphilipps]
azurerm_storage_account.mrgpjeanphilipps: Creating...
azurerm_storage_account.mrgpjeanphilipps: Still creating... [10s elapsed]
azurerm_storage_account.mrgpjeanphilipps: Still creating... [20s elapsed]
azurerm_storage_account.mrgpjeanphilipps: Still creating... [30s elapsed]
azurerm_storage_account.mrgpjeanphilipps: Still creating... [40s elapsed]
azurerm_storage_account.mrgpjeanphilipps: Still creating... [50s elapsed]
azurerm_storage_account.mrgpjeanphilipps: Still creating... [1m0s elapsed]
azurerm_storage_account.mrgpjeanphilipps: Still creating... [1m10s elapsed]
azurerm_storage_account.mrgpjeanphilipps: Creation complete after 1m17s [id=/subscriptions/8de89c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-tf-statefiles-jeanphilipps/providers/Microsoft.Storage/storageAccounts/tfstatesacctjpls]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:
storage_account_name = "tfstatesacctjpls"
PS C:\terraform\ResourceGroup> ]
```

[illegible]

The top screenshot shows the Microsoft Azure portal interface. The breadcrumb navigation indicates the path: Home > Resource groups > rg-tf-statefiles-jeanphilipps. The page title is 'Resource groups' with a search icon and a close button. Below the title, there are options to 'Create', 'Manage view', 'Refresh', 'Export to CSV', 'Open query', and 'Assign tags'. A search bar contains 'rg-tf-statefiles'. Below the search bar, there are filters for 'Subscription equals all' and 'Location equals all'. The results show one resource group: 'rg-tf-statefiles-jeanphilipps' located in 'East US' under the 'Azure for Students' subscription.

The bottom screenshot shows the 'tfstatesacctjpls' storage account page. The breadcrumb navigation indicates the path: Home > Resource groups > rg-tf-statefiles-jeanphilipps > tfstatesacctjpls. The page title is 'tfstatesacctjpls' with a search icon and a close button. Below the title, there are options to 'Upload', 'Open in Explorer', 'Delete', 'Move', 'Refresh', 'Open in mobile', 'CLI / PS', and 'Feedback'. The left sidebar shows the 'Overview' tab selected. The main content area displays the 'Essentials' section, which includes the 'Resource group' as 'rg-tf-statefiles-jeanphilipps', 'Location' as 'eastus', 'Subscription' as 'Azure for Students', 'Subscription ID' as '8de9c41-7afc-4788-9666-7d97d6c02cab', 'Disk state' as 'Available', and 'Tags' as 'environment: dev' and 'owner: IT'. The 'Performance' section shows 'Standard' performance, 'Locally-redundant storage (LRS)' replication, 'StorageV2 (general purpose v2)' account kind, and 'Succeeded' provisioning state. The 'Properties' section shows 'Blob service' settings, including 'Hierarchical namespace' (Disabled), 'Default access tier' (Hot), 'Blob anonymous access' (Enabled), 'Blob soft delete' (Disabled), 'Container soft delete' (Disabled), 'Versioning' (Disabled), 'Change feed' (Disabled), 'NFS v3' (Disabled), and 'Allow cross-tenant replication' (Disabled). The 'Security' section shows 'Require secure transfer for REST API operations' (Enabled), 'Storage account key access' (Enabled), 'Minimum TLS version' (Version 1.2), and 'Infrastructure encryption' (Disabled). The 'Networking' section shows 'Allow access from' (All networks), 'Private endpoint connections' (0), and 'Network routing' (Microsoft network routing).

- **Destroy resources provisioned with “terraform destroy”**

```

C:\terraform>terraform plan -destroy -var tfstatefiles_jeanphilipps
azure_resource_group.azrgjeanphilipps: Refreshing state... [id=/subscriptions/8de9c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-tf-statefiles-jeanphilipps]
azure_storage_account.azstgjeanphilipps: Refreshing state... [id=/subscriptions/8de9c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-tf-statefiles-jeanphilipps/providers/Microsoft.Storage/storageAccounts/tfstatesacctjpls]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy

Terraform will perform the following actions:

# azure_resource_group.azrgjeanphilipps will be destroyed
resource "azure_resource_group" "azrgjeanphilipps" {
  id      = "/subscriptions/8de9c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-tf-statefiles-jeanphilipps" -> null
  location = "eastus" -> null
  name     = "rg-tf-statefiles-jeanphilipps" -> null
  tags    = {} -> null
  # (4 unchanged attribute hidden)
}

# azure_storage_account.azstgjeanphilipps will be destroyed
resource "azure_storage_account" "azstgjeanphilipps" {
  access_tier           = "net" -> null
  account_kind          = "Storage" -> null
  account_replication_type = "LRS" -> null
  account_replication     = "Standard" -> null
  allow_nested_items_to_be_public = true -> null
  cross_tenant_replication_enabled = false -> null
  default_to_outh_authentication = false -> null
  dns_endpoint_type      = "Standard" -> null
  https_traffic_only_enabled = true -> null
  id                     = "/subscriptions/8de9c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-tf-statefiles-jeanphilipps/providers/Microsoft.Storage/storageAccounts/tfstatesacctjpls" -> null
  infrastructure_encryption_enabled = false -> null
  is_hns_enabled         = false -> null
  large_file_share_enabled = true -> null
  local_user_enabled     = true -> null
  location               = "eastus" -> null
  min_tls_version        = "TLS1_2" -> null
  name                   = "tfstatesacctjpls" -> null
  nfv1_enabled           = false -> null
  primary_access_key      = (sensitive value) -> null
  primary_blob_connection_string = (sensitive value) -> null
  primary_blob_endpoint    = "https://tfstatesacctjpls.blob.core.windows.net/" -> null
  primary_blob_host        = "tfstatesacctjpls.blob.core.windows.net" -> null
  primary_connection_string = (sensitive value) -> null
  primary_dfs_endpoint     = "https://tfstatesacctjpls.dfs.core.windows.net/" -> null
  primary_dfs_host         = "tfstatesacctjpls.dfs.core.windows.net" -> null
  primary_file_endpoint    = "https://tfstatesacctjpls.file.core.windows.net/" -> null
}

```

```

- blob_properties {
  - change_feed_enabled      = false -> null
  - change_feed_retention_in_days = 0 -> null
  - last_access_time_enabled   = false -> null
  - versioning_enabled        = false -> null
  # (1 unchanged attribute hidden)
}

- queue_properties {
  - hour_metrics {
    - enabled      = false -> null
    - include_apis = false -> null
    - retention_policy_days = 0 -> null
    - version      = "1.0" -> null
  }
  - logging {
    - delete      = false -> null
    - read        = false -> null
    - retention_policy_days = 0 -> null
    - version      = "1.0" -> null
    - write       = false -> null
  }
  - minute_metrics {
    - enabled      = false -> null
    - include_apis = false -> null
    - retention_policy_days = 0 -> null
    - version      = "1.0" -> null
  }
}

- share_properties {
  - retention_policy {
    - days = 7 -> null
  }
}
}

```

Plan: 0 to add, 0 to change, 2 to destroy.

Changes to Outputs:

- storage_account_name = "tfstatesacctjpls" -> null

Saved the plan to: tfplandestroy_jeanphilippels

To perform exactly these actions, run the following command to apply:

```
terraform apply "tfplandestroy_jeanphilippels"
```

PS C:\terraform\ResourceGroup>

Microsoft Azure Search resources, services, and docs (G+)

Home > Resource groups

Default Directory (jeanphilippel@gmail.com@microsoft.com)

+ Create Manage view Refresh Export to CSV Open query Assign tags

rg-tf-statefiles Subscription equals all Location equals all Add filter

Showing 0 to 0 of 0 records. No grouping List view

Name Subscription Location

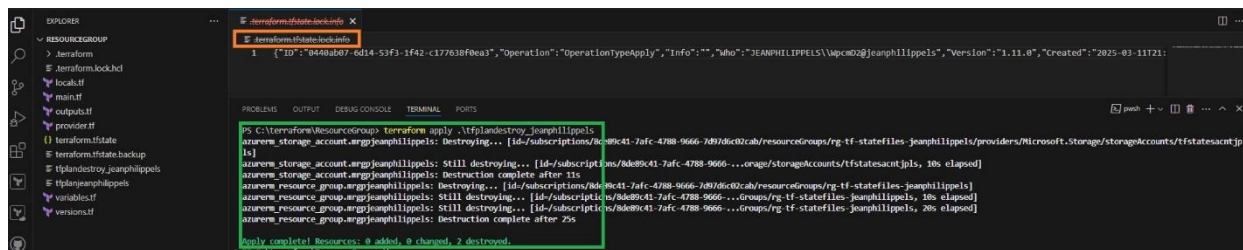
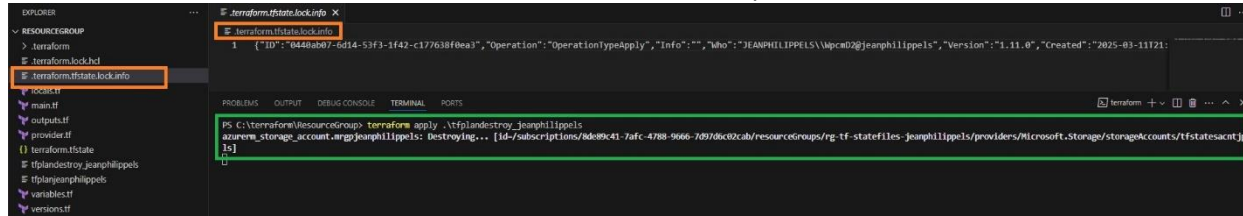
No resource groups match your filters

Resource groups provide a logical container to manage and organize Azure resources, simplifying administration and enabling efficient resource management.

+ Create Clear filters

Learn more

When a terraform apply is executing a `.terraform.tfstate.lock.info` is created in order to lock the tfstate so no one can execute at the same time a terraform command and corrupt the infrastructure.

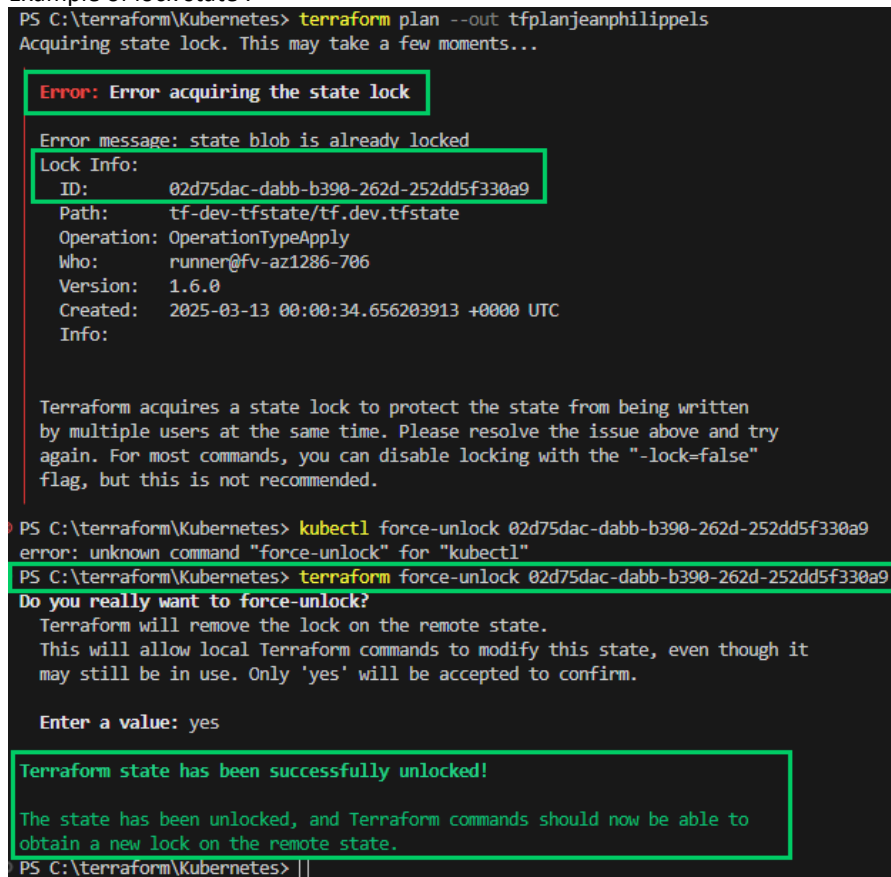


- **Configure terraform backend with Microsoft Azure**

State file can contain secrets, keeping it in Azure storage enhance security.

This will ensure that the latest version of the state file is used, while locking other user to deploy.

Example of lock state :



We provision the backend with terraform on Microsoft Azure.

- Provision storage account and container without backend block = terraform.tfstate locally

- After provisioning: add the backend block with references to the resources just created
- Run terraform init.
- Terraform will ask to migrate the local state file into Microsoft Azure Storage

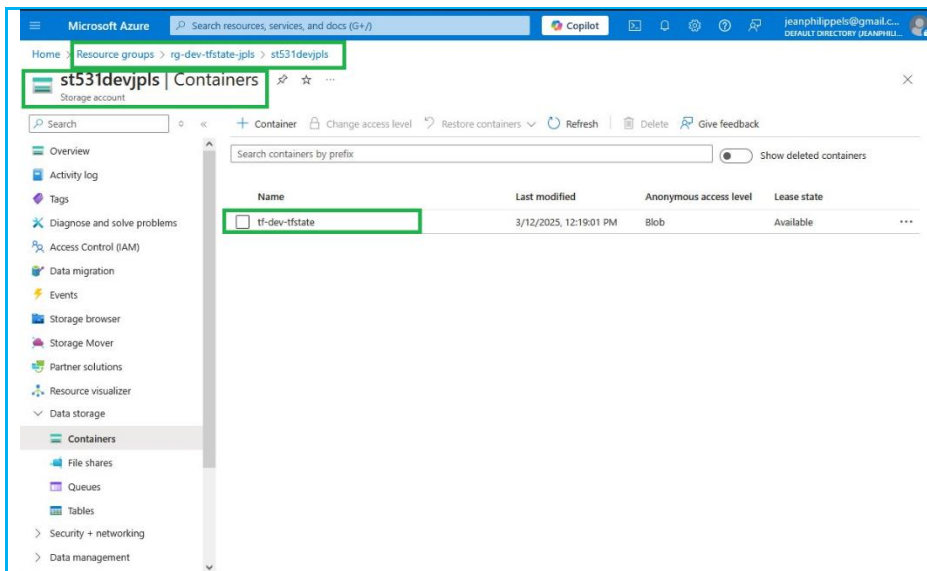
```
PS C:\terraform\ResourceGroup> terraform plan -out tfplanjeanphilipps

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

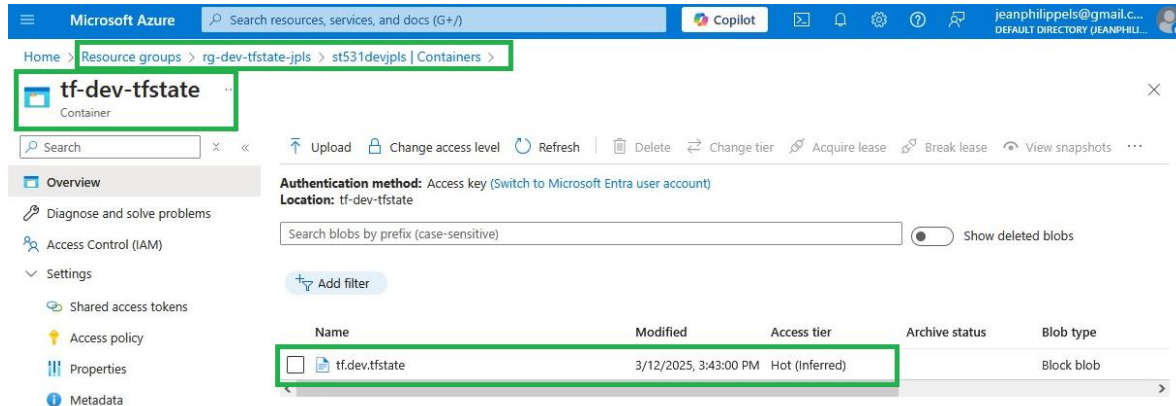
Terraform will perform the following actions:

# azure_rm_resource_group.mrgpjeanphilipps will be created
+ resource "azure_rm_resource_group" "mrgpjeanphilipps" {
+   id           = (known after apply)
+   location     = "eastus"
+   name        = "rg-dev-tfstate-jpls"
}

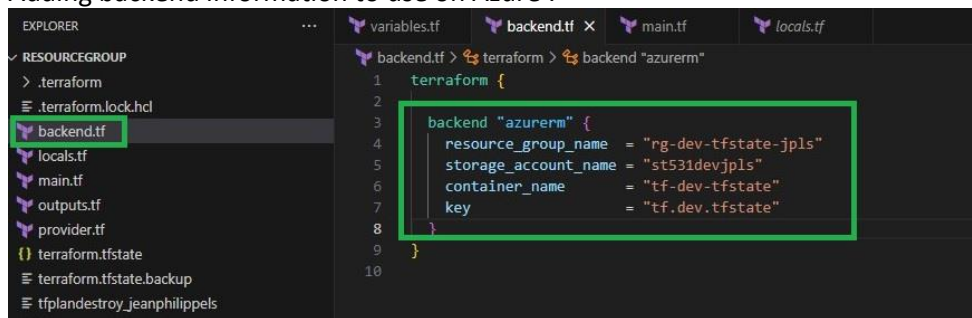
# azure_rm_storage_account.mrgpjeanphilipps will be created
+ resource "azure_rm_storage_account" "mrgpjeanphilipps" {
+   access_tier                = (known after apply)
+   account_kind               = "StorageV2"
+   account_replication_type   = "LRS"
+   account_tier               = "Standard"
+   allow_nested_items_to_be_public = true
+   cross_tenant_replication_enabled = false
+   default_to_oauth_authentication = false
+   dns_endpoint_type         = "Standard"
+   https_traffic_only_enabled = true
+   id                        = (known after apply)
+   infrastructure_encryption_enabled = false
+   is_hns_enabled            = false
+   large_file_share_enabled   = (known after apply)
+   local_user_enabled         = true
+   location                  = "eastus"
+   min_tls_version           = "TLS1_2"
+   name                      = "st53idevjpls"
+   nfsv3_enabled             = false
+   primary_access_key         = (sensitive value)
+   primary_blob_connection_string = (sensitive value)
+   primary_blob_endpoint      = (known after apply)
+   primary_blob_host          = (known after apply)
+   primary_blob_internet_endpoint = (known after apply)
+   primary_blob_internet_host = (known after apply)
+   primary_blob_microsoft_endpoint = (known after apply)
+   primary_blob_microsoft_host = (known after apply)
+   primary_connection_string   = (sensitive value)
+   primary_dfs_endpoint        = (known after apply)
+   primary_dfs_host            = (known after apply)
+   primary_dfs_internet_endpoint = (known after apply)
+   primary_dfs_internet_host = (known after apply)
+   primary_dfs_microsoft_endpoint = (known after apply)
+   primary_dfs_microsoft_host = (known after apply)
+   primary_file_endpoint       = (known after apply)
+   primary_file_host           = (known after apply)
+   primary_file_internet_endpoint = (known after apply)
+   primary_file_internet_host = (known after apply)
}
```

Tf State file is stored in the Azure container with the "tf.dev.tfstate" name.



Adding backend information to use on Azure :



The screenshot shows the Visual Studio Code interface with the Explorer pane on the left displaying the project structure. The main editor shows the `backend.tf` file with the following content:

```

1 terraform {
2
3   backend "azurerm" {
4     resource_group_name = "rg-dev-tfstate-jpls"
5     storage_account_name = "st531devjpls"
6     container_name      = "tf-dev-tfstate"
7     key                  = "tf.dev.tfstate"
8   }
9 }
10

```

The TERMINAL pane at the bottom shows the output of the `terraform init` command:

```

PS C:\terraform\ResourceGroup> terraform init
Initializing the backend...
Acquiring state lock. This may take a few moments...
Do you want to copy existing state to the new backend?
Pre-existing state was found while migrating the previous "local" backend to the
newly configured "azurerm" backend. No existing state was found in the newly
configured "azurerm" backend. Do you want to copy this state to the new "azurerm"
backend? Enter "yes" to copy and "no" to start with an empty state.
Enter a value: yes
Releasing state lock. This may take a few moments...
Successfully configured the backend "azurerm"! Terraform will automatically
use this backend unless the backend configuration changes.
Initializing provider plugins...
- Reusing previous version of hashicorp/azurerm from the dependency lock file
- Using previously-installed hashicorp/azurerm v4.8.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\terraform\ResourceGroup>

```

To verify the setup, the local instance of **terraform.tfstate** file has been deleted .
We run command `terraform state list`. The output is as follow:

The screenshot shows the Visual Studio Code interface with the Explorer pane on the left displaying the project structure. The main editor shows the `backend.tf` file with the following content:

```

1 terraform {
2
3   backend "azurerm" {
4     resource_group_name = "rg-dev-tfstate-jpls"
5     storage_account_name = "st531devjpls"
6     container_name      = "tf-dev-tfstate"
7     key                  = "tf.dev.tfstate"
8   }
9 }
10

```

The TERMINAL pane at the bottom shows the output of the `terraform state list` command:

```

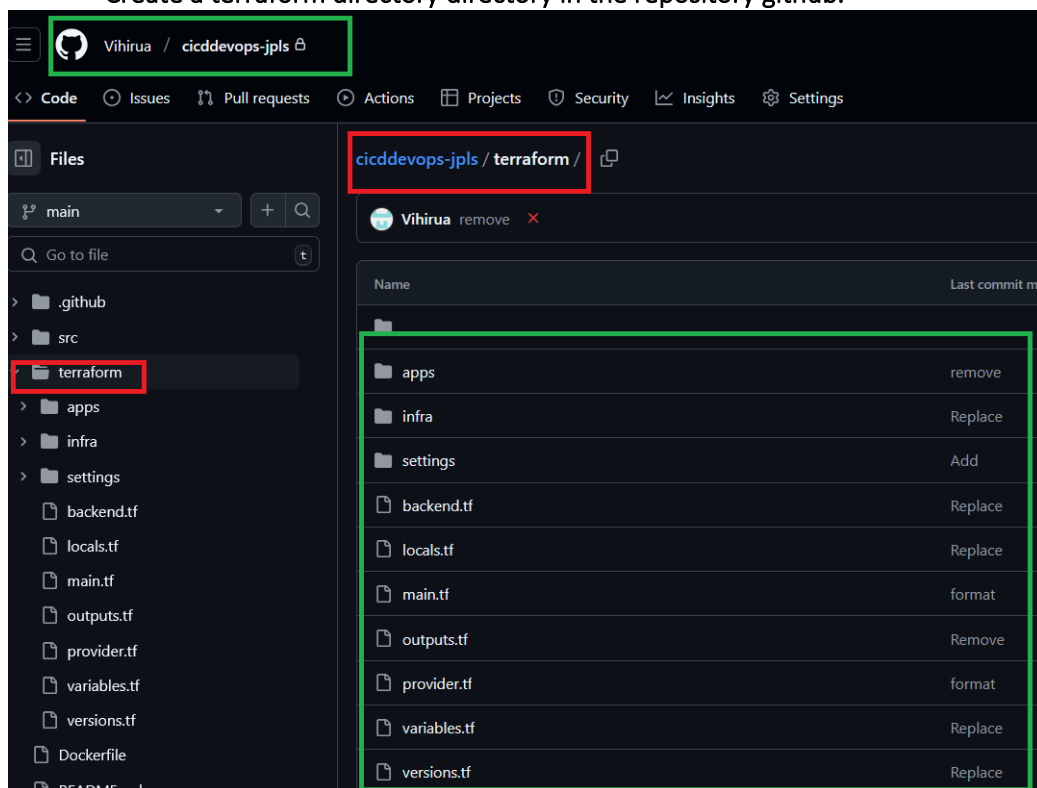
PS C:\terraform\ResourceGroup> terraform state list
azurerm_resource_group.mrgpjeanphilippels
azurerm_storage_account.mrgpjeanphilippels
azurerm_storage_container.terraformbackend
PS C:\terraform\ResourceGroup>

```

Deployment in Azure Kubernetes Cluster through Terraform

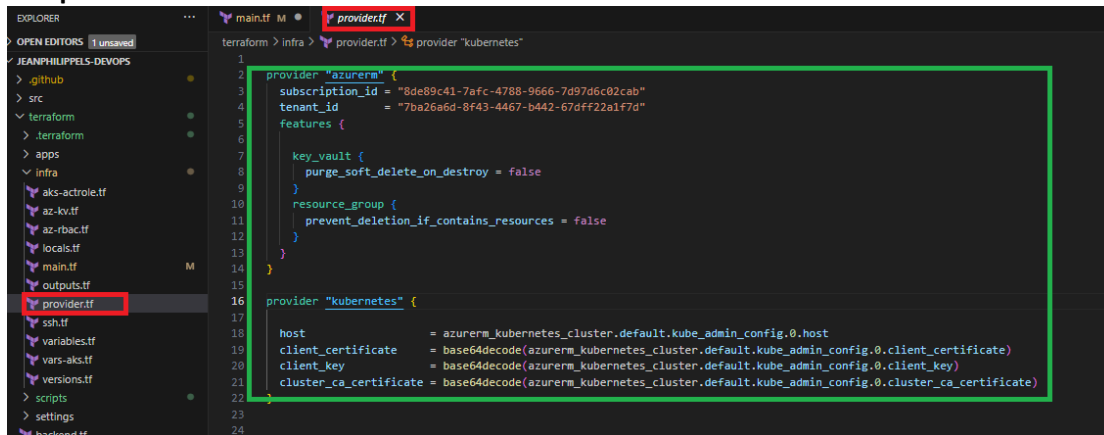
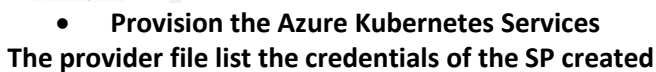
- **Deploy the container using Kubernetes / Apply Kubernetes Configuration**
 - Create a terraform directory in the repository github.
 - *Configure Azure Kubernetes Services*
- *Provision the Azure Kubernetes Services*
- *Create the custom namespace for devops applications*
- *Create the Secrets allowing to use private docker repository*
- *Create the Service account*
- *Create the role cluster for the service account*
- *Bind the role cluster to the service account*
- *Create the custom devops web application deployment with secret from a docker hub repository*
- *Create the service custom devops web application deployment*
- *Provision the Apps in Azure Kubernetes Services*
- *Testing the devops web deployment*

- *Create a terraform directory in the repository github.*

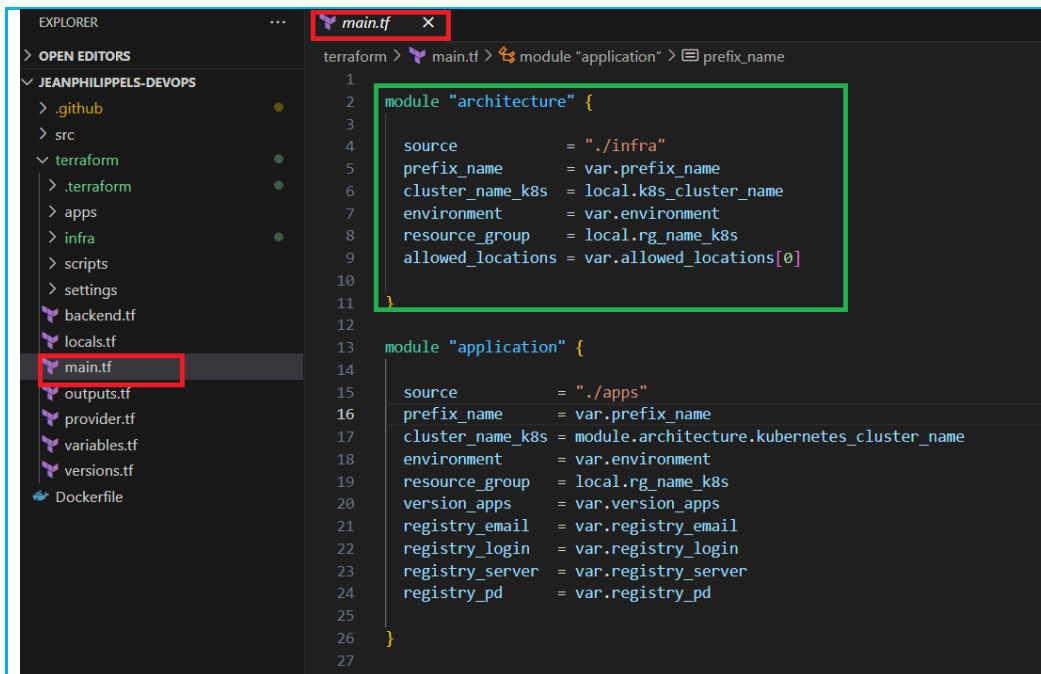


- *Configure Azure Kubernetes Services*

In order terraform to access Microsoft Azure, a Service Principal Name has to be created. Contributor Role is giving to the Service Principal Name.



Project : DevOps implementation Document | version 1.0 | Final Release



- **Module architecture:**

The variables of the module are the parameters that need to be fill up when calling in the main root directory of terraform :


```

module.architecture.azure_rm_kubernetes_cluster.default: Still creating... [5m30s elapsed]
module.architecture.azure_rm_kubernetes_cluster.default: Creation complete after 5m36s [id=/subscriptions/8de89c41-7afc-4788-9666-7d97d6c02cab/resourceGroups/rg-dev-k8s-jpls/providers/Microsoft.ContainerService/managedClusters/k8s-dev-cluster]
module.architecture.azuread_group_member.groupmemberspaks_msi: Creating...
module.architecture.kubernetes_cluster_role_binding.act_bind: Creating...
module.architecture.local_file.kubeconfig: Creating...
module.architecture.kubernetes_cluster_role.actallow: Creating...
module.architecture.local_file.kubeconfig: Creation complete after 0s [id=4ed9bf0358a7c541e55958f1176c8306ce1307c4]
module.architecture.azuread_group_member.groupmemberspaks_msi: Creation complete after 1s [id=436fa8a1-831a-4ebe-8618-7afa59ab06e3/member/a210dbbd-2f7e-47c1-83fb-8db8359adbca]
module.architecture.kubernetes_cluster_role.actallow: Creation complete after 1s [id=managed-cluster]
module.architecture.kubernetes_cluster_role_binding.act_bind: Creation complete after 1s [id=managed-cluster-bind]

Warning: Resource targeting is in effect

You are creating a plan with the -target option, which means that the result of this plan may not represent all of the changes requested by the current configuration.

The -target option is not for routine use, and is provided only for exceptional situations such as recovering from errors or mistakes, or when Terraform specifically suggests to use it as part of an error message.

Warning: Applied changes may be incomplete

The plan was created with the -target option in effect, so some changes requested in the configuration may have been ignored and the output values may not be fully updated. Run the following command to verify that no other changes are pending:

    terraform plan

Note that the -target option is not suitable for routine use, and is provided only for exceptional situations such as recovering from errors or mistakes, or when Terraform specifically suggests to use it as part of an error message.

Releasing state lock. This may take a few moments...

Apply complete! Resources: 22 added, 0 changed, 0 destroyed.

Outputs:
aks_msi_principal_id = "a210dbbd-2f7e-47c1-83fb-8db8359adbca"
client_certificate = <sensitive>
client_key = <sensitive>
cluster_ca_certificate = <sensitive>
cluster_name = "k8s-dev-cluster"
cluster_username = <sensitive>
host = <sensitive>
kube_config = <sensitive>

```

k8s-dev-cluster

Kubernetes service

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Microsoft Defender for Cloud (preview)

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Node pools

Kubernetes versions

Node sizes

Networking

API server address

Network configuration

Pod CIDR

Service CIDR

DNS service IP

Cilium dataplane

Network Policy

Azure Kubernetes Services relies on Virtual Machine Scale Set which a group of load balanced Virtual Machines.

aks-k8sdevnode-32010145-vmss
Virtual machine scale set

Overview

Essentials

Resource group (move) : [rg-dev-k8s-jpls-nrg](#)
 Status : **2 out of 2 succeeded**
 Location : East US
 Subscription (move) : [Azure for Students](#)
 Subscription ID : 8de89c41-7afc-4788-9666-7d97d6c02cab

Operating system : Linux
 Size : Standard_D2_v2 (2 instances)
 Public IP address : [172.171.57.60](#)
 Public address (IPv6) : -
 Virtual network/subnet : [aks-vnet-12231162/aks-subnet](#)
 Orchestration mode : Uniform
 Time created : 3/14/2025, 10:24 PM UTC

Tags (edit) : aks-managed-azure-cni-overlay : true aks-managed-consolidated-additio... : e77fda75-0122-11f0-9f7b-c...

Properties Monitoring Capabilities (6) Recommendations Tutorials

Virtual machine profile
 Operating system : Linux
 Capacity reservation group : -
 Hibernation : Disabled

Azure Spot
 Azure Spot : Disabled

Availability + scaling
 Availability zone : -
 Proximity placement group : -
 Colocation status : -

Status
 Provisioning state : **Succeeded**
 Power state : 2 out of 2 running

Networking
 Public IP address : -
 Public IP address (IPv6) : -
 Virtual network/subnet : [aks-vnet-12231162/aks-subnet](#)

Size
 Size : Standard_D2_v2
 vCPUs : 2

rg-dev-k8s-jpls-nrg
Resource group

Resources

Filter for any field... Type equals all Location equals all Add filter

Showing 1 to 7 of 7 records. Show hidden types No grouping List view

Name	Type	Location
416ac108-8245-403b-bbb7-ce34abc7dc80	Public IP address	East US
aks-agentpool-12231162-nsg	Network security group	East US
aks-k8sdevnode-34073874-vmss	Virtual machine scale set	East US
aks-vnet-12231162	Virtual network	East US
azurepolicy-k8s-dev-cluster	Managed Identity	East US
k8s-dev-cluster-agentpool	Managed Identity	East US
kubernetes	Load balancer	East US

- **Create the custom namespace for devops applications**

The screenshot displays a development environment with two main components: a code editor and the Microsoft Azure portal.

Code Editor (Top): The file explorer on the left shows a project structure for 'JEANPHILIPPELS-DEVOPS'. The 'aks-apps.tf' file is selected and highlighted with a red box. The main editor shows the Terraform configuration for a Kubernetes namespace, with the resource definition highlighted by a green box:

```

1 # ----- AKS APPLICATIONS ----- #
2
3 #Kubernetes namespace to hold application
4 resource "kubernetes_namespace" "tf_k8s_ns" {
5
6   metadata {
7     name = local.k8s_namespace_devops_apps
8     labels = {
9       environment = local.environment_tags["environment"]
10      managed_by   = local.default_tags["managed_by"]
11      type         = local.default_tags["type"]
12      department   = local.default_tags["department"]
13    }
14  }
15 }
16
17 # ----- Secrets ----- #
18

```

Azure Portal (Bottom): The portal shows the 'k8s-dev-cluster | Namespaces' page. The 'Namespaces' tab is selected and highlighted with a red box. A table lists the namespaces, with the 'dev-apps-jeanphilipels' namespace highlighted by a green box:

Name	Status	Age
kube-system	Active	48 minutes
kube-public	Active	48 minutes
kube-node-lease	Active	48 minutes
default	Active	48 minutes
gatekeeper-system	Active	47 minutes
dev-apps-jeanphilipels	Active	44 minutes

Below the screenshot, there is a bullet point indicating the next step:

- **Create the Secrets allowing to use private docker repository**

The screenshot shows the Terraform code for `aks-apps.tf`. The left sidebar shows the file explorer with `aks-apps.tf` selected. The main editor displays the following code:

```

1 # ----- AKS APPLICATIONS ----- #
2
3 #Kubernetes namespace to hold application
4 resource "kubernetes_namespace" "tf_k8s_ns" {
5
6   metadata {
7     name = local.k8s_namespace_devops_apps
8     labels = {
9       environment = local.environment_tags["environment"]
10      managed_by   = local.default_tags["managed_by"]
11      type         = local.default_tags["type"]
12      department   = local.default_tags["department"]
13    }
14  }
15 }
16
17 # ----- Secrets ----- #
18 resource "kubernetes_secret" "adsec" {
19   metadata {
20     name       = local.k8s_secrets_docker
21     namespace = kubernetes_namespace.tf_k8s_ns.metadata[0].name
22   }
23   type = "kubernetes.io/dockerconfigjson"
24
25   data = {
26     ".dockerconfigjson" = jsonencode({
27       auths = {
28         "${local.registry_server}" = {
29           "username" = local.registry_username
30           "password" = local.registry_pd
31           "email"    = local.registry_email
32           "auth"     = base64encode("${local.registry_username}:${local.registry_pd}")
33         }
34       }
35     })
36   }
37 }
38
39
40

```

○ **Create the Service account**

The screenshot shows the Terraform code for `aks-role.tf`. The left sidebar shows the file explorer with `aks-role.tf` selected. The main editor displays the following code:

```

1 terraform > apps > aks-role.tf > resource "kubernetes_cluster_role" "allow_deploy" > rule > [ ] api_groups
2
3 # ----- Service account ----- #
4 resource "kubernetes_service_account" "sa_apps" {
5   metadata {
6     name       = local.k8s_sa
7     namespace = local.k8s_namespace_devops_apps
8   }
9   image_pull_secret {
10     name = kubernetes_secret.adsec.metadata[0].name
11   }
12 }
13
14 # ----- Cluster Role ----- #
15 resource "kubernetes_cluster_role" "allow_deploy" {
16   metadata {
17     name = var.k8s_role_cr_name
18   }
19 }
20
21 rule {

```

○ **Create the role cluster for the service account**


```

terraform > apps > aks-role.tf > resource "kubernetes_cluster_role" "allow_deploy" > rule > [ ] api_groups
4 resource "kubernetes_service_account" "sa_apps" {
13 }
14
15 # ----- Cluster Role ----- #
16 resource "kubernetes_cluster_role" "allow_deploy" {
17   metadata {
18     name = var.k8s_role_cr_name
19   }
20
21   rule {
22     api_groups = [""]
23     resources = ["pods"]
24     verbs      = ["list", "get", "watch", "create", "delete"]
25   }
26
27   rule {
28     api_groups = [""]
29     resources = ["pods/exec"]
30     verbs      = ["create"]
31   }
32
33   rule {
34     api_groups = [""]
35     resources = ["pods/log"]
36     verbs      = ["get"]
37   }
38
39   rule {
40     api_groups = [""]
41     resources = ["pods/attach"]
42     verbs      = ["list", "get", "create", "delete", "update"]
43   }
44
45   rule {
46     api_groups = [""]
47     resources = ["secrets"]
48     verbs      = ["list", "get", "create", "delete", "update"]
49   }
50
51   rule {
52     api_groups = [""]
53     resources = ["configmaps"]
54     verbs      = ["list", "get", "create", "delete", "update"]
55   }
56
57   rule {
58     api_groups = [""]
59     resources = ["services"]
60     verbs      = ["list", "get", "create", "delete", "update", "patch"]
61   }
62
63   rule {
64     api_groups = ["apps"]
65     resources = ["deployments"]
66     verbs      = ["list", "get", "create", "delete", "update", "patch"]
67   }
68 }
69

```

○ **Bind the role cluster to the service account**

```

terraform > apps > aks-role.tf > resource "kubernetes_cluster_role_binding" "allow_deploy" > subject > name
16 resource "kubernetes_cluster_role" "allow_deploy" {
51   rule {
52     api_groups = [""]
53     resources = ["services"]
54     verbs      = ["list", "get", "create", "delete", "update", "patch"]
55   }
56
57   rule {
58     api_groups = ["apps"]
59     resources = ["deployments"]
60     verbs      = ["list", "get", "create", "delete", "update", "patch"]
61   }
62 }
63
64 # ----- Cluster Role Binding ----- #
65 resource "kubernetes_cluster_role_binding" "allow_deploy" {
66   metadata {
67     name = local.k8s_role_cr_name_binding
68   }
69
70   role_ref {
71     api_group = "rbac.authorization.k8s.io"
72     kind      = "ClusterRole"
73     name      = kubernetes_cluster_role.allow_deploy.metadata.0.name
74   }
75
76   subject {
77     kind      = "ServiceAccount"
78     name      = kubernetes_service_account.sa_apps.metadata.0.name
79     namespace = kubernetes_namespace.tf_k8s_ns.metadata[0].name
80   }
81 }
82

```

- *Create the custom devops web application deployment with secret from a docker hub repository*

```

77
78
79 #Kubernetes deployment
80
81 resource "kubernetes_deployment" "appsk8s" {
82   depends_on = [
83     kubernetes_namespace.tf_k8s_ns,
84     kubernetes_service_account.sa_apps,
85     kubernetes_secret.adsec,
86     kubernetes_cluster_role.allow_deploy,
87     kubernetes_cluster_role_binding.allow_deploy, kubernetes_service.apps:8s
88   ]
89   metadata {
90     name      = local.k8s_deploy_name
91     namespace = kubernetes_namespace.tf_k8s_ns.metadata[0].name
92     labels = {
93       app      = "${local.k8s_label_name}"
94       version  = "${var.version_apps}"
95       environment = local.environment_tags["environment"]
96       managed_by = local.default_tags["managed_by"]
97       type      = local.default_tags["type"]
98       department = local.default_tags["department"]
99     }
100   }
101   spec {
102     replicas = var.k8s_replicas
103     selector {
104       match_labels = {
105         app      = "${local.k8s_label_name}"
106         version  = "${var.version_apps}"
107         environment = local.environment_tags["environment"]
108         managed_by = local.default_tags["managed_by"]
109         type      = local.default_tags["type"]
110         department = local.default_tags["department"]
111       }
112     }
113   }
114   template {
115     metadata {
116       labels = {
117         app      = "${local.k8s_label_name}"
118         version  = "${var.version_apps}"
119         environment = local.environment_tags["environment"]
120         managed_by = local.default_tags["managed_by"]
121         type      = local.default_tags["type"]
122         department = local.default_tags["department"]
123       }
124     }
125   }
126   spec {
127     service_account_name = local.k8s_sa
128   }
129 }

```

```

terraform > apps > aks-apps.tf > resource "kubernetes_deployment" "appsk8s" > spec > template > spec
81 resource "kubernetes_deployment" "appsk8s" {
102   spec {
115     template {
127       spec {
129         container {
135           path = "/"
136           port = var.k8s_container_port
137         }
138
139         initial_delay_seconds = 10
140         period_seconds = 5
141         failure_threshold = 3
142       }
143     }
144   }
145
146   liveness_probe {
147     http_get {
148       path = "/"
149       port = var.k8s_container_port
150     }
151
152     initial_delay_seconds = 10
153     period_seconds = 5
154     failure_threshold = 3
155   }
156
157   resources {
158     limits = {
159       cpu = "20m"
160       memory = "80Mi"
161     }
162     requests = {
163       cpu = "10m"
164       memory = "20Mi"
165     }
166   }
167
168   port {
169     container_port = var.k8s_container_port
170   }
171
172   image_pull_secrets {
173     name = kubernetes_secret.adsec.metadata[0].name
174   }
175 }
176
177 }
178

```

- **Create the service custom devops web application deployment**

```

terraform > apps > aks-apps.tf > resource "kubernetes_deployment" "appsk8s" > spec
25 resource "kubernetes_secret" "adsec" {
44 }
45 }
46
47
48
49 #Kubernetes service to access the Apps webpage
50 resource "kubernetes_service" "appsk8s" {
51   /*
52     depends_on = [
53       kubernetes_namespace.tf_k8s_ns, kubernetes_service_account.sa_apps, kubernetes_secret.adsec
54     ]
55   */
56   metadata {
57     name = local.k8s_pods_name
58     namespace = kubernetes_namespace.tf_k8s_ns.metadata[0].name
59   }
60   spec {
61     selector = {
62       app = "${local.k8s_label_name}"
63       version = "${var.version_apps}"
64       environment = local.environment_tags["environment"]
65       managed_by = local.default_tags["managed_by"]
66       type = local.default_tags["type"]
67       department = local.default_tags["department"]
68     }
69     port {
70       port = var.k8s_svc_port
71       target_port = var.k8s_svc_des_port
72     }
73     type = var.k8s_svc_type
74   }
75 }
76
77

```


Plan: 8 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```
+ apps_service_ip = (known after apply)
module.application.kubernetes_namespace.tf_k8s_ns: Creating...
module.application.kubernetes_cluster_role.allow_deploy: Creating...
module.application.kubernetes_cluster_role.allow_deploy: Creation complete after 1s [id=CR-DEV-DEPLOY]
module.application.kubernetes_namespace.tf_k8s_ns: Creation complete after 1s [id=dev-apps-jeanphilippels]
module.application.kubernetes_secret.adsec: Creating...
module.application.kubernetes_service.appsk8s: Creating...
module.application.kubernetes_secret.adsec: Creation complete after 1s [id=dev-apps-jeanphilippels/devops-dev-sec]
module.application.kubernetes_service_account.sa_apps: Creating...
module.application.kubernetes_service_account.sa_apps: Creation complete after 0s [id=dev-apps-jeanphilippels/devopsdevsa]
module.application.kubernetes_cluster_role_binding.allow_deploy: Creating...
module.application.kubernetes_cluster_role_binding.allow_deploy: Creation complete after 1s [id=CR-BIND-dev-DEPLOY]
module.application.kubernetes_service.appsk8s: Creation complete after 9s [id=dev-apps-jeanphilippels/devopsdevpods]
module.application.kubernetes_deployment.appsk8s: Creating...
module.application.kubernetes_deployment.appsk8s: Creation complete after 1s [id=dev-apps-jeanphilippels/devops-deploy-jeanphilippels]
module.application.kubernetes_horizontal_pod_autoscaler.hpaajpls: Creating...
module.application.kubernetes_horizontal_pod_autoscaler.hpaajpls: Creation complete after 1s [id=dev-apps-jeanphilippels/k8s-hpa-dev-jeanphilippels]
```

Warning: Resource targeting is in effect

You are creating a plan with the `-target` option, which means that the result of this plan may not represent all of the changes requested by the current configuration.

The `-target` option is not for routine use, and is provided only for exceptional situations such as recovering from errors or mistakes, or when Terraform specifically suggests to use it as part of an error message.

Warning: Applied changes may be incomplete

The plan was created with the `-target` option in effect, so some changes requested in the configuration may have been ignored and the output values may not be fully updated. Run the following command to verify that no other changes are pending:

```
terraform plan
```

Note that the `-target` option is not suitable for routine use, and is provided only for exceptional situations such as recovering from errors or mistakes, or when Terraform specifically suggests to use it as part of an error message.

Releasing state lock. This may take a few moments...

Apply complete! Resources: 8 added, 0 changed, 0 destroyed.

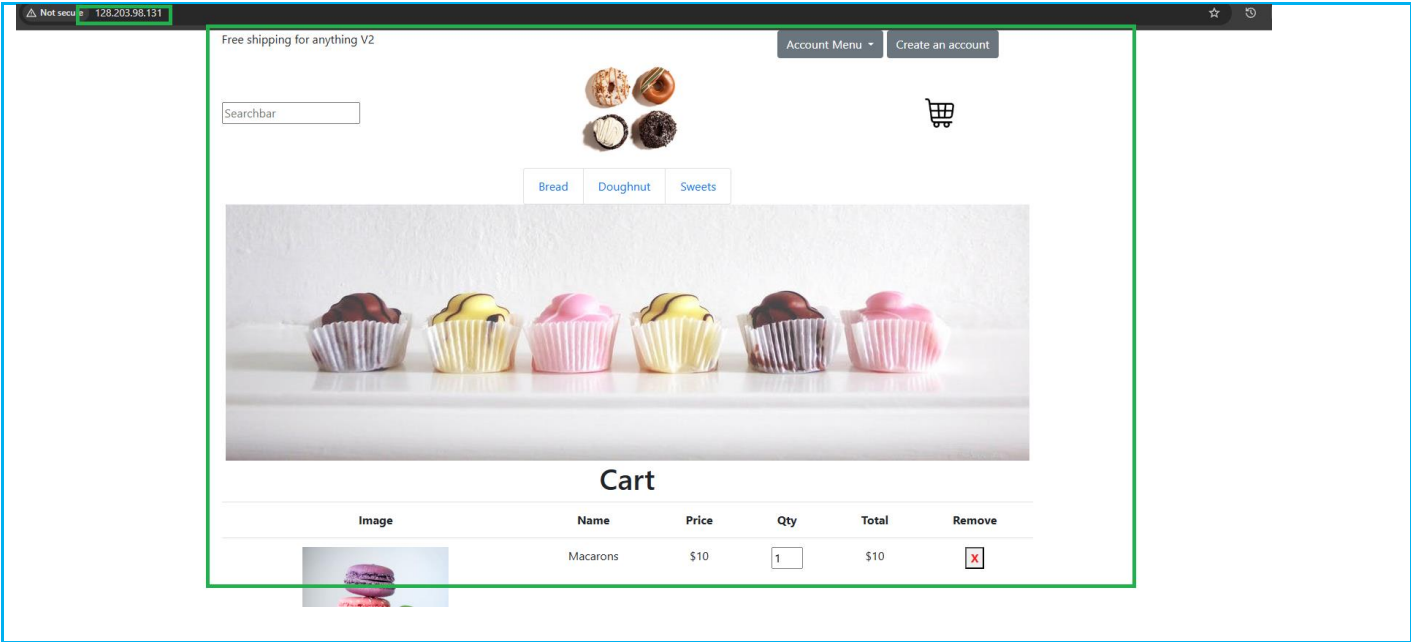
- **Testing the devops web deployment**

```
PS C:\terraform_jpls\Kubernetes_jpls> kubectl get deploy -n dev-apps-jeanphilippels
NAME                                READY    UP-TO-DATE    AVAILABLE    AGE
devops-deploy-jeanphilippels        1/1      1              1             4m19s
PS C:\terraform_jpls\Kubernetes_jpls> kubectl get svc -n dev-apps-jeanphilippels
NAME            TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
devopsdevpods   LoadBalancer 10.0.121.108   128.203.98.131 80:32748/TCP     29m
PS C:\terraform_jpls\Kubernetes_jpls>
```

```

PS C:\terraform_jpls\Kubernetes_jpls> kubectl describe deploy/devops-deploy-jeanphilippels -n dev-apps-jeanphilippels
Name:                devops-deploy-jeanphilippels
Namespace:           dev-apps-jeanphilippels
CreationTimestamp:    Sun, 16 Mar 2025 15:18:35 +1300
Labels:              app=devops_dev_apps
                    department=devops
                    environment=dev
                    managed_by=terraform
                    version=V1.0
Annotations:         deployment.kubernetes.io/revision: 1
Selector:             app=devops_dev_apps,department=devops,environment=dev,managed_by=terraform,version=V1.0
Replicas:             1 desired | 1 updated | 1 total | 1 available | 0 unavailable
StrategyType:        RollingUpdate
MinReadySeconds:      0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
Pod Template:
  Labels:             app=devops_dev_apps
                    department=devops
                    environment=dev
                    managed_by=terraform
                    version=V1.0
  Service Account:    devopsdevsa
  Containers:
    devops-dev-apps:
      Image:           docker1sp123/devops-jeanphilippels:V1.0
      Port:            80/TCP
      Host Port:       0/TCP
      Limits:
        cpu:           20m
        memory:        80Mi
      Requests:
        cpu:           10m
        memory:        20Mi
      Liveness:         http-get http://:80/ delay=10s timeout=1s period=5s #success=1 #failure=3
      Readiness:        http-get http://:80/ delay=10s timeout=1s period=5s #success=1 #failure=3
      Environment:      <none>
      Mounts:           <none>
      Volumes:          <none>
  Node-Selectors:      <none>
  Tolerations:         <none>
Conditions:
  Type            Status  Reason
  ----            -
  Available        True    MinimumReplicasAvailable
  Progressing      True    NewReplicaSetAvailable
OldReplicaSets:    <none>
NewReplicaSet:     devops-deploy-jeanphilippels-8657f6847d (1/1 replicas created)
Events:
  Type    Reason             Age    From                      Message
  ----    -
  Normal  ScalingReplicaSet  4m56s  deployment-controller     Scaled up replica set devops-deploy-jeanphilippels-8657f6847d from 0 to 1
PS C:\terraform_jpls\Kubernetes_jpls>

```



SETTING UP CI/CD PIPELINE

GitHub Actions | Workflow Push Docker image to private repository

- Building DevOps image with Dockerfile
- Building container with the DevOps image
- Testing container with the DevOps image
- Pushing DevOps image to private docker repository

GitHub Actions is a platform that allows to do Continuous Integration and Continuous Delivery (CI/ CD) by automating build, test and deployment in pipeline

• Building DevOps image with Dockerfile

```
github > workflows > tf-docker.yml
1  name: 2-Build and Push docker image
2  description: 'Build and push docker image'
3  on:
4    workflow_run:
5      workflows: ["1-Build and Test"]
6      types: [completed]
7      branches:
8        - 'dev'
9    #workflow_dispatch:
10
11  env:
12    #ENV: "${{ vars.ENV }}"
13    ENV: "dev"
14    version: "V1.0"
15    container_name: "mycaketest-jpls"
16
17  jobs:
18    release-docker:
19      name: Buid | Push Images to docker hub
20      runs-on: ubuntu-latest
21      steps:
22        - name: Check out the repo
23          uses: actions/checkout@v3
24          with:
25            ref: 'dev'
26
27        - name: Build Docker Image
28          run: docker build -t ${{ secrets.DOCKERHUB_USERNAME }}/${{ secrets.REPOSITORY }}:${{ env.version }} .
29
```

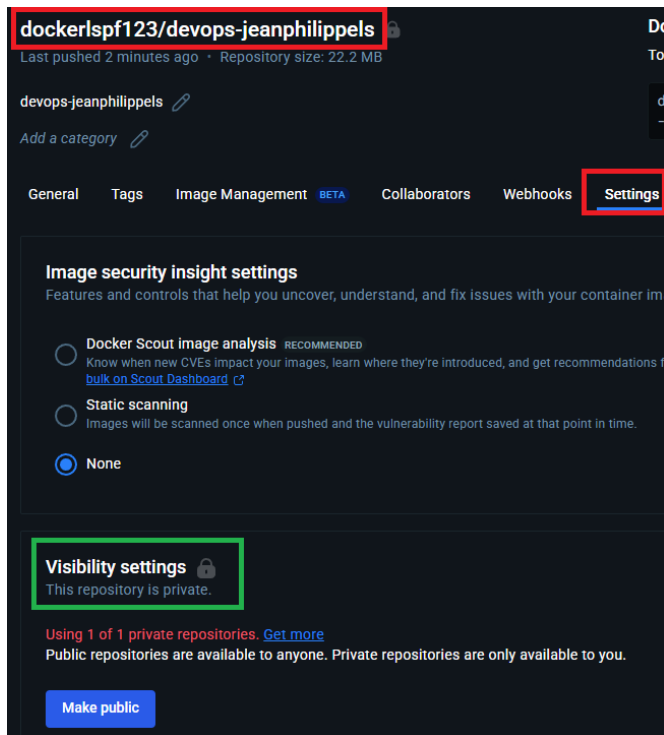
• Building container with the DevOps image

```
29
30  - name: Run Docker Container
31    run: docker run -d -p 80:80 --name $container_name ${{ secrets.DOCKERHUB_USERNAME }}/${{ secrets.REPOSITORY }}:${{ env.version }}
32
```

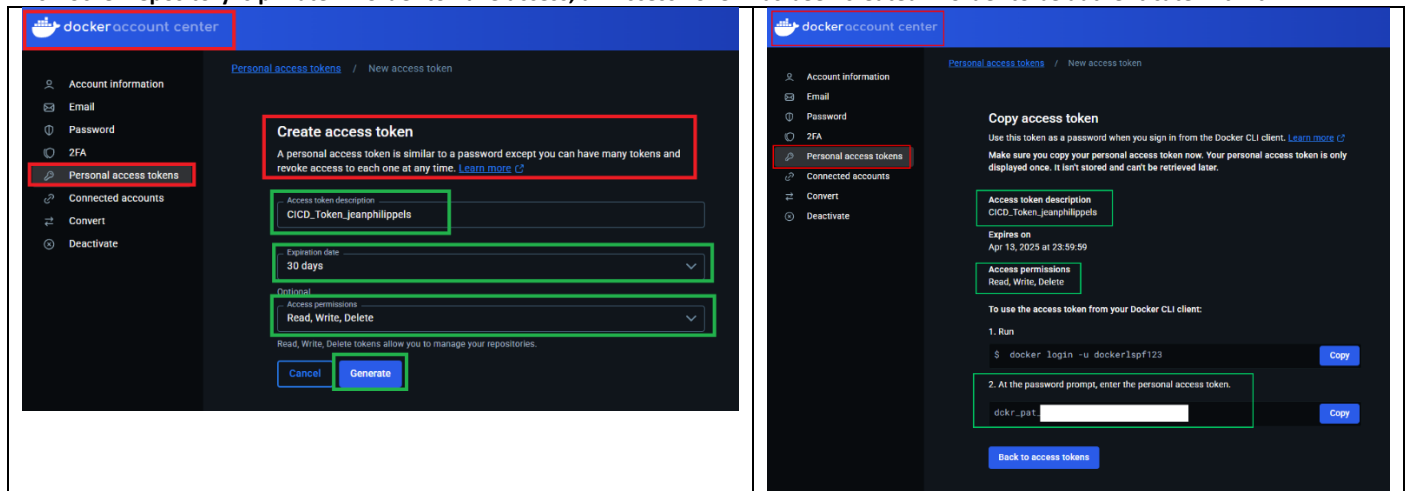
• Testing container with the DevOps image

```
33  - name: test Index Page
34    run: |
35      sleep 10 # Allow container to start
36      curl --fail http://localhost:80/
37
```

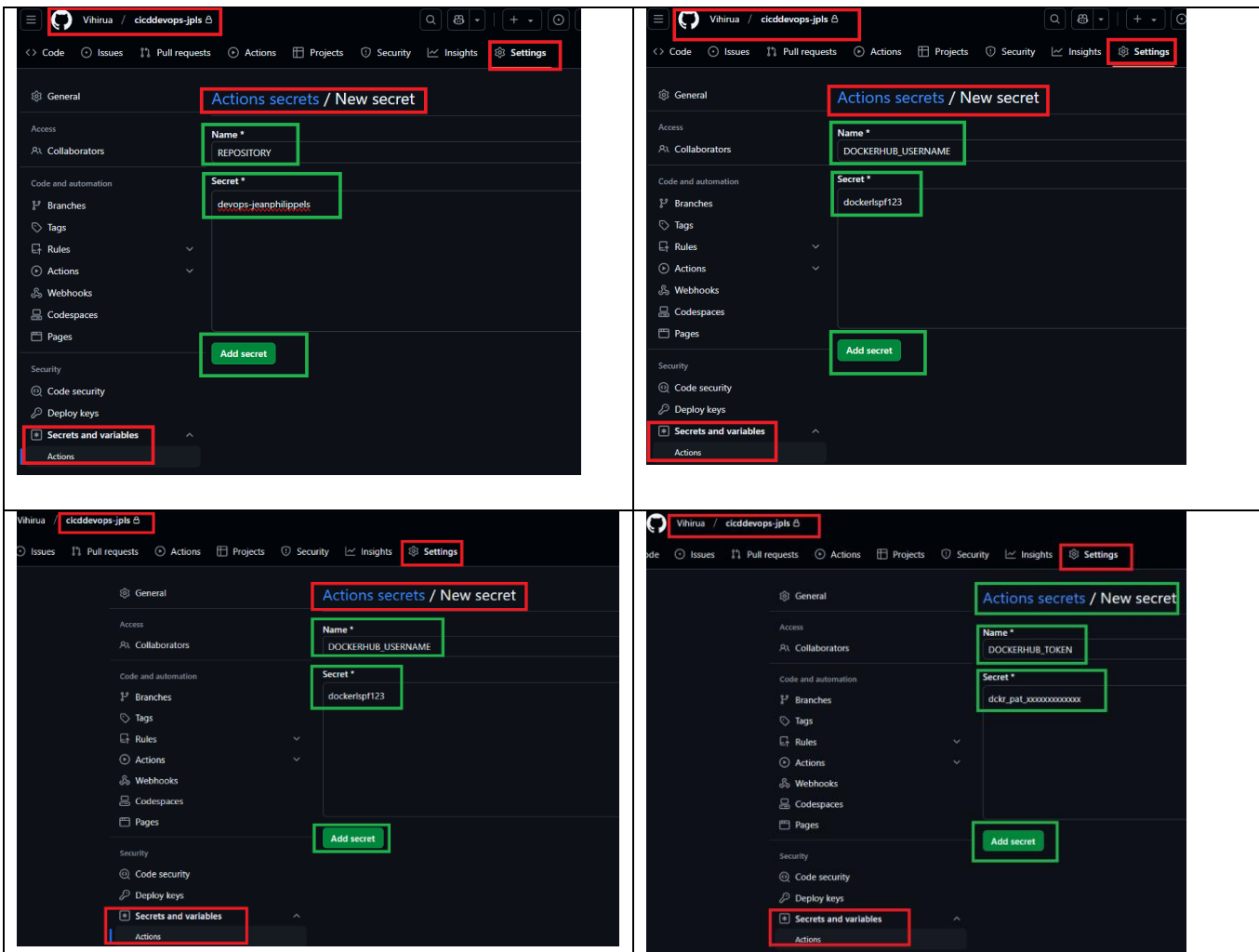

- Pushing images to private docker repository



This Docker repository is private. In order to have access, an Access Token has been created in order to be authenticate with it .



Secrets have been created in GitHub to use for GitHub actions:



Workflow on GitHub for Building and Pushing Docker image into the private docker hub registry :

```

github > workflows > %_ tf-docker.yml
1  name: 2-Build and Push docker image
2  description: 'Build and push docker image'
3  on:
4    workflow_run:
5      workflows: ["1-Build and Test"]
6      types: [completed]
7      branches:
8        - 'dev'
9  #workflow_dispatch:
10
11 env:
12   #ENV: "${{ vars.ENV }}"
13   ENV: "dev"
14   version: "v1.0"
15   container_name: "mycaketest-jpls"
16
17 jobs:
18   release-docker:
19     name: Build | Push Images to docker hub
20     runs-on: ubuntu-latest
21     steps:
22       - name: Check out the repo
23         uses: actions/checkout@v3
24         with:
25           ref: 'dev'
26
27       - name: Build Docker Image
28         run: docker build -t ${{ secrets.DOCKERHUB_USERNAME }}/${{ secrets.REPOSITORY }}:${{ env.version }} .
29
30       - name: Run Docker Container
31         run: docker run -d -p 80:80 --name $container_name ${{ secrets.DOCKERHUB_USERNAME }}/${{ secrets.REPOSITORY }}:${{ env.version }}
32
33       - name: Test Index Page
34         run: |
35           sleep 10 # Allow container to start
36           curl --fail http://localhost:80/
37
38       - name: Stop Test Container
39         run: |
40           sleep 5 # Allow container to stop
41           docker stop $(docker ps -q)
42
43       - name: Delete Test Container
44         run: |
45           sleep 5 # Allow container to be deleted
46           docker rm $(docker ps -a -q)
47
48       - name: Setup Docker
49         uses: docker/setup-buildx-action@v2
50
51       - name: Set Docker Credentials
52         uses: docker/login-action@v2
53         with:
54           username: ${{ secrets.DOCKERHUB_USERNAME }}
55           password: ${{ secrets.DOCKERHUB_TOKEN }}
56
57       - name: Build Image and Push
58         id: build-push-api
59         uses: docker/build-push-action@v3
60         with:
61           context: .
62           file: Dockerfile
63           push: true
64           tags: ${{ secrets.DOCKERHUB_USERNAME }}/${{ secrets.REPOSITORY }}:${{ env.version }}
65
66

```

A commit into the repository launched the workflow :

The screenshot shows the GitHub Actions interface for the repository 'Vihirua / cicdddevops-jpls'. The workflow '2-Build and Push docker image' (run #72) is highlighted with a green box. The 'Jobs' section shows 'Buid | Push Images to docker hub' as the active job, also highlighted with a green box. The 'Run details' section lists the steps of the job, which are also highlighted with a green box:

- > Set up job
- > Check out the repo
- > Build Docker Image
- > Run Docker Container
- > Test Index Page
- > Stop Test Container
- > Delete Test Container
- > Setup Docker
- > Set Docker Credentials
- > Build Image and Push
- > Post Build Image and Push
- > Post Set Docker Credentials
- > Post Setup Docker
- > Post Check out the repo
- > Complete job

After the workflow has been successfully ran, the new image has been push into the private docker repository :

The screenshot shows the Docker Hub repository page for 'dockerlsp123/devops-jeanphilippels'. The repository name is highlighted with a red box. The 'Tags' tab is selected and highlighted with a red box. The 'Tags' section shows a list of tags, with 'V1.0' highlighted by a green box. The 'Digest' and 'OS/ARCH' columns are also highlighted with a green box.

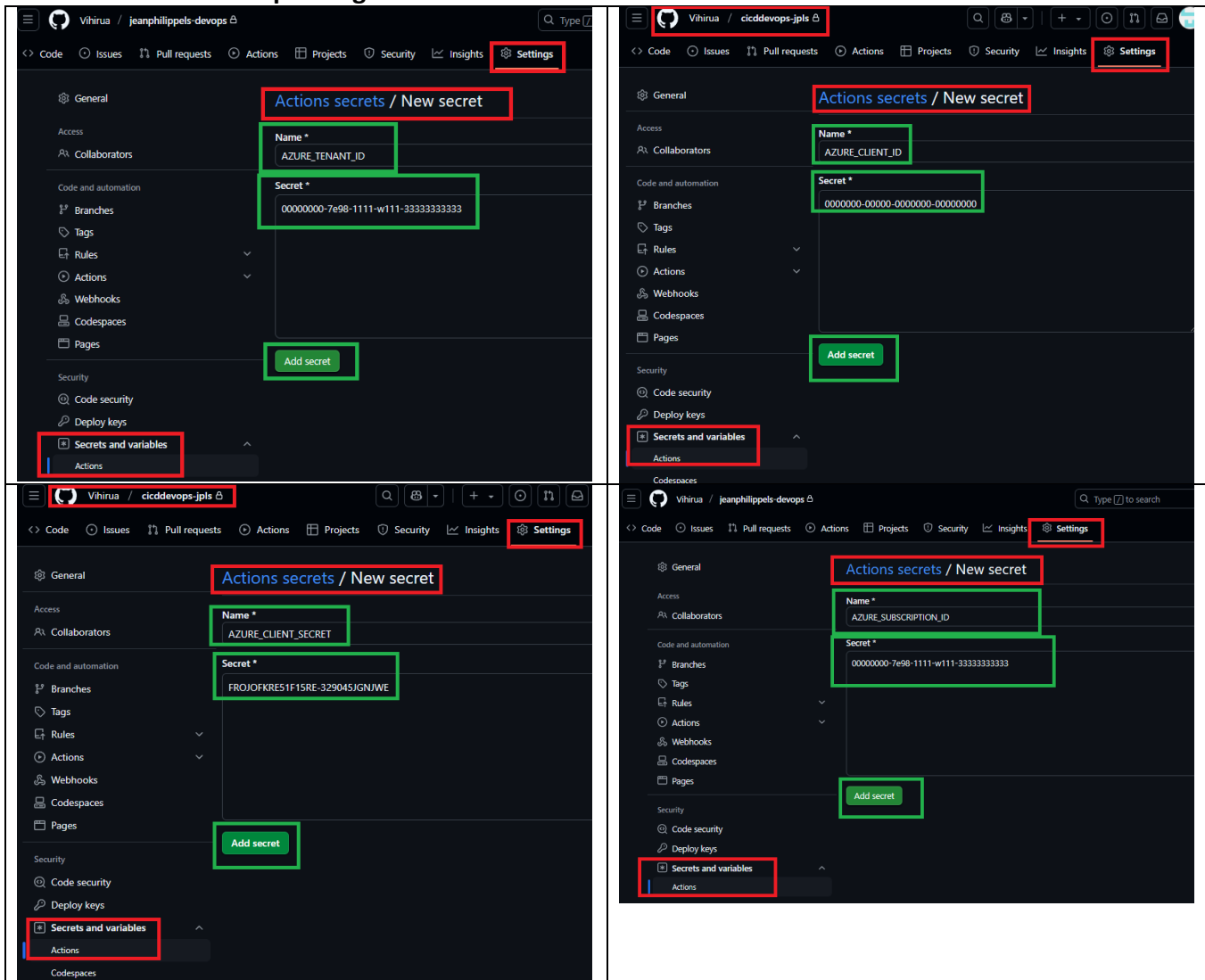
Digest	OS/ARCH	Last pull	Compressed size
6ee15c449e28	linux/amd64	less than 1 day	20.17 MB

GitHub Actions | Workflow Terraform

Workflow Unit Test

When pushing code to the branch, this workflow will analyze if the scripts are well formed and respond to the format.

- **Test unit when pushing into branches**



```

.github / workflows > ⌘ tf-build_test.yml
1  name: 1-UnitTest
2  description: 'Build and Test code'
3  on:
4    push:
5      branches:
6        - 'dev'
7      paths:
8        - 'terraform/**'
9        - '.github/workflows/tf-build_test.yml'
10 concurrency:
11   group: ${{ github.workflow }}-${{ github.ref }} # Grouping to ensure only one job per branch
12   cancel-in-progress: true
13 env:
14   ARM_CLIENT_ID: ${{ secrets.AZURE_CLIENT_ID }}
15   ARM_CLIENT_SECRET: ${{ secrets.AZURE_CLIENT_SECRET }}
16   ARM_SUBSCRIPTION_ID: ${{ secrets.AZURE_SUBSCRIPTION_ID }}
17   ARM_TENANT_ID: ${{ secrets.AZURE_TENANT_ID }}
18   #ENV: "${{ vars.ENV }}"
19   ENV: "dev"
20   CONF_FILE: "backend-unittest.conf"
21   BACKEND_SCRIPT: "backend-unit-test"
22   BACKEND_RM_SCRIPT: "backend-unit-test-rm"
23   WORKING_DIR: "./terraform"
24   WORKING_SCRIPTS_DIR: "scripts"
25 jobs:
26   terraform-unit-tests:
27     name: "Terraform Unit Tests"
28     runs-on: ubuntu-latest
29     concurrency:
30       group: dev_environment
31       cancel-in-progress: true
32     environment: development
33     defaults:
34       run:
35         shell: bash
36         working-directory: ${{ env.WORKING_DIR }}
37     steps:
38       # Checkout the repository
39       - name: Checkout
40         uses: actions/checkout@v3
41       - name: Initialization of the Backend on Azure
42         id: init-backend
43         shell: bash
44         run: |
45           chmod +x ./${{ env.WORKING_SCRIPTS_DIR }}/${{ env.BACKEND_SCRIPT }}.sh
46           ./${{ env.WORKING_SCRIPTS_DIR }}/${{ env.BACKEND_SCRIPT }}.sh
47       env:
48         AZURE_CLIENT_ID: ${{ secrets.AZURE_CLIENT_ID }}
49         AZURE_CLIENT_SECRET: ${{ secrets.AZURE_CLIENT_SECRET }}
50         AZURE_TENANT_ID: ${{ secrets.AZURE_TENANT_ID }}
51       # Test all Terraform files the latest version

```

```

51 # Install Terraform - the latest version
52 - name: Set up Terraform
53   uses: hashicorp/setup-terraform@v3
54   with:
55     terraform_version: 1.6.0 # Specify Terraform version
56 # Initialize Terraform
57 - name: Terraform Init
58   id: tf-init
59   run: terraform init -reconfigure -backend-config=settings/${ENV}/${CONF_FILE}
60 - name: Destroy Backend on Azure
61   id: destroy-backend
62   shell: bash
63   run: |
64     chmod +x ./${{ env.WORKING_SCRIPTS_DIR }}/${{ env.BACKEND_RM_SCRIPT }}.sh
65     ./${{ env.WORKING_SCRIPTS_DIR }}/${{ env.BACKEND_RM_SCRIPT }}.sh
66   env:
67     AZURE_CLIENT_ID: ${ secrets.AZURE_CLIENT_ID }
68     AZURE_CLIENT_SECRET: ${ secrets.AZURE_CLIENT_SECRET }
69     AZURE_TENANT_ID: ${ secrets.AZURE_TENANT_ID }
70 # Validate terraform files
71 - name: Terraform Validate
72   run: terraform validate
73
74 # Checks that all Terraform configuration files adhere to a canonical format
75 - name: Terraform Format
76   run: terraform fmt -check -recursive
77
78

```

1-Build and Test

tf-build_test.yml

Filter workflow runs



Help us improve GitHub Actions

Tell us how to make GitHub Actions work better for you with three quick questions.

Give feedback



101 workflow runs

Event

Status

Branch

Actor

✓ Add

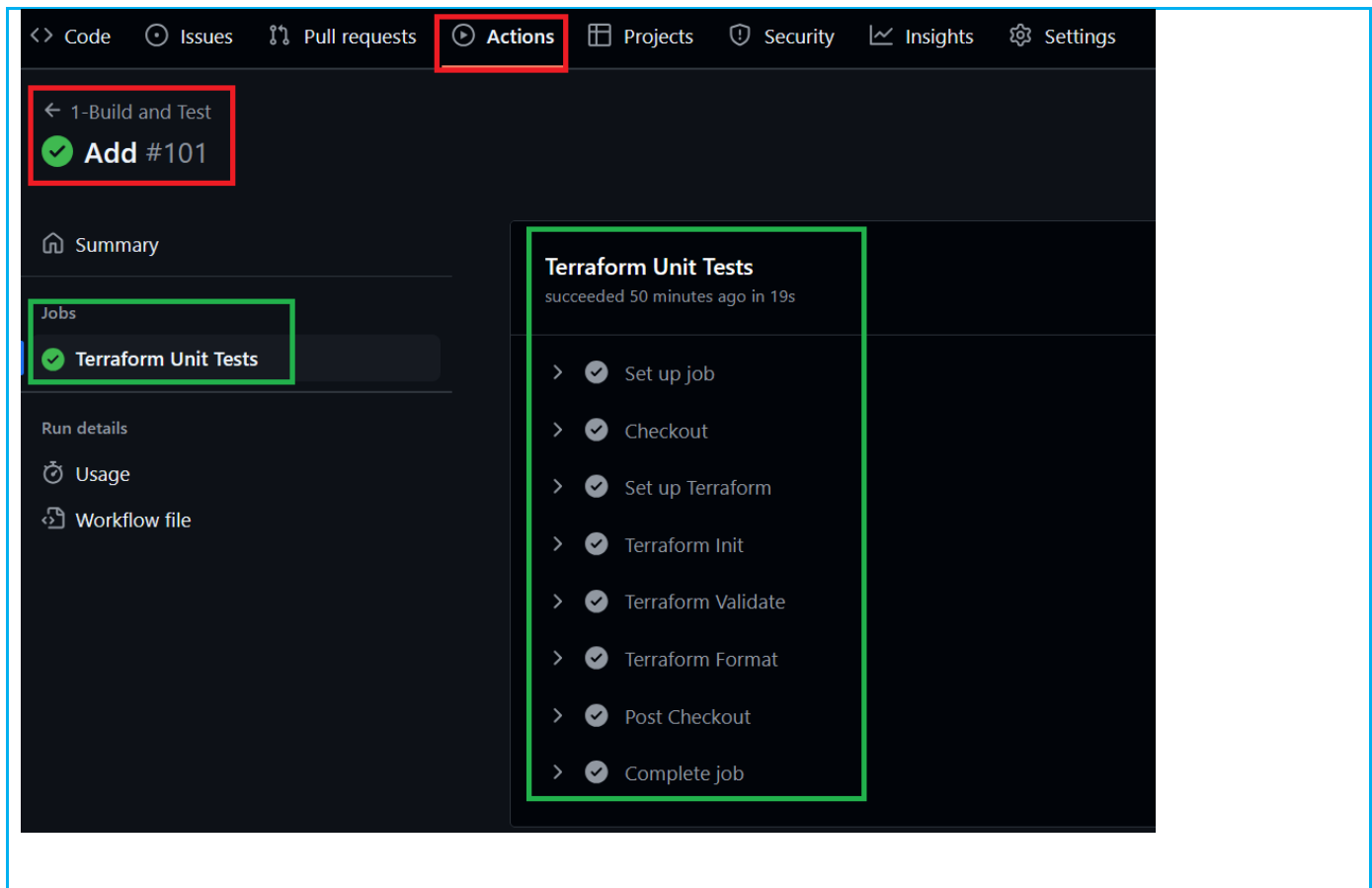
1-Build and Test #101: Commit a8f34b0 pushed by Vihirua

dev

51 minutes ago

25s





Workflow Init Backend

This workflow provisions the backend to use on Azure allowing to test manually module with terraform if teams want to .

- *Shell to create resource group and container for the backend*

The screenshot shows the GitHub Actions interface for a workflow named '9-BackEnd-Init' in the repository 'jeanphilippels-devops'. The 'Actions' tab is selected. The workflow run '#4' is shown as successful. The 'Backend Init' job is highlighted, showing a list of steps: 'Set up job', 'Checkout', 'Initialization of the Backend on Azure' (highlighted with a green box), 'Post Checkout', and 'Complete job'. The 'Initialization of the Backend on Azure' step is marked as successful.

The screenshot shows the Microsoft Azure portal 'Resource groups' page. The search bar contains 'rg-dev-tfstate'. The table below shows the results:

Name ↑	Subscription	Location
rg-dev-tfstate-jpls	Azure for Students	East US

Workflow Azure Kubernetes Services

This workflow provisions Azure Kubernetes Services with the backend and the Azure Kubernetes Services

The screenshot displays the GitHub Actions interface for the repository 'jeanphilippes-devops'. The 'Actions' tab is selected, showing a workflow named '3-Provision AKS' with the file 'tf-publish-todev-nod.yml'. The workflow has 32 runs. A 'Run workflow' button is highlighted in green. A dropdown menu is open, showing 'Use workflow from' with 'Branch: dev' selected and a 'Run workflow' button highlighted in green.

The second screenshot shows the details of the '3-Provision AKS #32' workflow run. The run is successful, triggered manually 18 hours ago by user 'Vihirua'. The summary shows a total duration of 7m 28s, a billable time of 9m, and 1 artifact. The workflow steps are listed on the left: Backend Initialization, Terraform Plan, and Terraform Apply. The workflow graph shows the sequence: Backend Initialization (1m 1s) → Terraform Plan (29s) → Terraform Apply.

Summary

Jobs	Status	Total duration	Billable time	Artifacts
- Backend Initialization - Terraform Plan - Terraform Apply	Success	7m 28s	9m	1

tf-publish-todev-nod.yml
on: workflow_dispatch

```

graph LR
    A[Backend Initialization 1m 1s] --> B[Terraform Plan 29s]
    B --> C[Terraform Apply 5m 43s]
  
```


Terraform Plan summary

Terraform Plan Output

▼ Click to expand

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create
<- read (data resources)

Terraform will perform the following actions:

# module.architecture.data.azuread_group.aksgroup will be read during apply
# (depends on a resource or a module with changes pending)
<- data "azuread_group" "aksgroup" {
  # <redacted> (known after apply)
  + auto_subscribe_new_members = (known after apply)
  + behaviors                  = (known after apply)
  + description                = (known after apply)
}
  
```

Summary

Jobs

- Backend Initialization
- Terraform Plan
- Terraform Apply

Run details

- Usage
- Workflow file

```

# module.architecture.local_file.kubeconfig will be created
+ resource "local_file" "kubeconfig" {
+   content           = (sensitive value)
+   content_base64sha256 = (known after apply)
+   content_base64sha512 = (known after apply)
+   content_md5       = (known after apply)
+   content_sha1      = (known after apply)
+   content_sha256    = (known after apply)
+   content_sha512    = (known after apply)
+   directory_permission = "0777"
+   file_permission    = "0777"
+   filename           = "../kubeconfig"
+   id                  = (known after apply)
}

# module.architecture.random_pet.ssh_key_name will be created
+ resource "random_pet" "ssh_key_name" {
+   id      = (known after apply)
+   length = 2
+   prefix = "ssh"
}

Plan: 16 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ aks_msi_principal_id = (known after apply)
+ client_certificate    = (sensitive value)
+ client_key            = (sensitive value)
+ cluster_ca_certificate = (sensitive value)
+ cluster_name          = "k8s-dev-cluster"
+ cluster_username      = (sensitive value)
+ host                  = (sensitive value)
+ kube_config            = (sensitive value)

```

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot



jeanphilippels@gmail.c...
DEFAULT DIRECTORY (JEANPHILI...

Home >

Resource groups

...

Default Directory (jeanphilippels@gmail.onmicrosoft.com)

 Manage view

 Refresh

 Export to CSV

 Open query

|

 Assign tags

 Group by none

 You are viewing a new version of Browse experience. Some features may be missing. [Click here to access the old experience.](#)

 rg-dev-k8s-jpls

Subscription equals all

Location equals all

 Add filter

☐

Name 1

Subscription

Location

☐

 rg-dev-k8s-jpls

...

Azure for Students

East US

☐

 rg-dev-k8s-jpls-nrg

...

Azure for Students

East US

The image displays two screenshots of the Microsoft Azure portal interface, showing resource groups and their associated resources.

Top Screenshot: Resource group 'rg-dev-k8s-jpls'

- The breadcrumb navigation shows 'Home > Resource groups > rg-dev-k8s-jpls'.
- The 'Essentials' section shows 'Resources' with a filter for 'k8s-'. The results show 1 record: 'k8s-dev-cluster' (Kubernetes service, East US).

Bottom Screenshot: Resource group 'rg-dev-k8s-jpls-nrg'

- The breadcrumb navigation shows 'Home > Resource groups > rg-dev-k8s-jpls-nrg'.
- The 'Essentials' section shows 'Resources' with a filter for 'Filter for any field...'. The results show 7 records:

Name	Type	Location
416ac108-8245-403b-bbb7-ce34abc7dc80	Public IP address	East US
aks-agentpool-12231162-nsg	Network security group	East US
aks-k8sdevnode-34073874-vmss	Virtual machine scale set	East US
aks-vnet-12231162	Virtual network	East US
azurepolicy-k8s-dev-cluster	Managed Identity	East US
k8s-dev-cluster-agentpool	Managed Identity	East US
kubernetes	Load balancer	East US

Workflow Applications to Azure Kubernetes Services

This workflow provisions the applications into the Azure Kubernetes Services

The screenshot shows the GitHub Actions interface for a workflow named '4-Provision APPS | AKS'. The workflow is highlighted with a red box. The 'Actions' tab is selected in the top navigation bar. The workflow run is shown as a success, triggered manually 18 hours ago, with a status of 'Success', a total duration of 2m 51s, a billable time of 4m, and 1 artifact. The workflow steps are listed on the left: Terraform Plan and Terraform Apply. The workflow file is 'tf-publish-todev-apps.yml' on the 'workflow_dispatch' event. A green box highlights the workflow steps: Terraform Plan (34s) and Terraform Apply (2m 6s).

The screenshot shows the Terraform Plan output for the workflow. The workflow run is highlighted with a red box. The 'Summary' tab is selected. The workflow steps are listed on the left: Terraform Plan and Terraform Apply. The workflow file is 'tf-publish-todev-apps.yml' on the 'workflow_dispatch' event. A green box highlights the Terraform Plan summary and output. The Terraform Plan summary shows the plan was successful. The Terraform Plan output shows the plan was successful and the following actions will be performed:

```
# module.application.kubernetes_cluster_role.allow_deploy will be created
+ resource "kubernetes_cluster_role" "allow_deploy" {
  + id = (known after apply)
```

The image displays two screenshots related to a DevOps implementation. The top screenshot shows the output of a Terraform plan command, highlighting the configuration for a Kubernetes service account. The bottom screenshot shows the GitHub Actions interface for a workflow named '4-Provision APPS | AKS', with the 'Terraform Plan' job highlighted.

Terraform Plan Output (Top Screenshot):

```
+ environment = "dev"
+ "managed_by" = "terraform"
+ "type" = "apps"
+ "version" = "v1.0"
}
+ session_affinity = "None"
+ type = "LoadBalancer"
}
+ port {
+ node_port = (known after apply)
+ port = 80
+ protocol = "TCP"
+ target_port = "80"
}
}
}

# module.application.kubernetes_service_account.sa_apps will be created
+ resource "kubernetes_service_account" "sa_apps" {
+ automount_service_account_token = true
+ default_secret_name = (known after apply)
+ id = (known after apply)
+ image_pull_secret {
+ name = "devops-dev-sec"
}
+ metadata {
+ generation = (known after apply)
+ name = "devopsdevsa"
+ namespace = "dev-apps-jeanphilippels"
+ resource_version = (known after apply)
+ uid = (known after apply)
}
}

Plan: 8 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ apps_service_ip = (known after apply)
```

GitHub Actions Interface (Bottom Screenshot):

The interface shows the workflow '4-Provision APPS | AKS' with a green checkmark indicating success. The 'Terraform Plan' job is highlighted, showing a list of steps:

- Set up job
- Checkout the repository to the GitHub Action runner
- Set Version of the Apps
- Set up Terraform
- Terraform Init
- Terraform Refresh
- Terraform format
- Terraform plan
- Publish Terraform Plan Applications
- Create String Output
- Publish Terraform Plan to Task Summary
- Post Checkout the repository to the GitHub Action runner
- Complete job

The screenshot displays the GitHub Actions interface for a repository named 'jeanphilippels-devops'. The 'Actions' tab is selected in the top navigation bar. The workflow '4-Provision APPS | AKS' is highlighted in the left sidebar, with its 'Terraform Apply' job also selected. The main panel shows the 'Terraform Apply' job details, indicating it 'succeeded 17 hours ago in 2m 6s'. A list of 18 steps is shown, all of which are completed, as indicated by green checkmarks. The steps include setting up the job, checkout, setting up Terraform, Terraform Init, Terraform Refresh, downloading the Terraform Plan, setting up az login, saving the tf state, running Terraform Apply, getting Terraform Output, showing the output variable, setting up kubectl, setting up kubelogin, setting the AKS context, and finally running 'Get Deploy' and 'Get Services'.

GitHub Actions workflow for '4-Provision APPS | AKS' (AKS #47). The workflow is successful, having completed 17 hours ago in 2m 6s.

Terraform Apply (succeeded 17 hours ago in 2m 6s)

Jobs:

- Terraform Plan
- Terraform Apply**

Run details:

- Usage
- Workflow file

Steps:

- Set up job
- Checkout
- Setup Terraform
- Terraform Init
- Terraform Refresh
- Download Terraform Plan
- Setup az login
- Save tf state
- Terraform Apply
- Get Terraform Output
- Showing Output Var
- Setup kubectl
- Setup kubelogin
- Set AKS context
- Run Get Deploy
- Run Get Services

Workflow Destroy

This workflow destroys provisioned azure resources

← 10-Terraform Destroy

✓ 10-Terraform Destroy #21

Summary

Jobs

- ✓ Terraform Plan
- ✓ Terraform Apply
- ✓ Backend Destroy

Run details

- Usage
- Workflow file

Manually triggered 19 hours ago

Status: Success

Total duration: 14m 1s

Billable time: 15m

Artifacts: 1

tf-destroy.yml
on: workflow_dispatch

Workflow steps:

- ✓ Terraform Plan (3m 40s)
- ✓ Terraform Apply (8m 38s)
- ✓ Backend Destroy (1m 27s)

Summary

Jobs

- ✓ Terraform Plan
- ✓ Terraform Apply
- ✓ Backend Destroy

Run details

- Usage
- Workflow file

```

      fullnameOverride: "sq-dev-jpls"
      monitoringPasscode: "L92F#503dmkPFNM"
      postgresql:
        enabled: true
    EOT,
  ] -> null
- verify          = false -> null
- version         = "2025.1.1" -> null
- wait            = true -> null
- wait_for_jobs   = false -> null
}

# module.sec.kubernetes_namespace.terraform_k8s_sq will be destroyed
- resource "kubernetes_namespace" "terraform_k8s_sq" {
  - id              = "sq-dev-jpls" -> null
  - wait_for_default_service_account = false -> null

  - metadata {
    - annotations = {} -> null
    - generation  = 0 -> null
    - labels      = {
      - "department" = "devops"
      - "environment" = "dev"
      - "managed_by" = "terraform"
      - "type" = "sec"
    } -> null
    - name          = "sq-dev-jpls" -> null
    - resource_version = "4481" -> null
    - uid            = "1641c7d9-4694-48c6-80c4-5f116a6e46b3" -> null
  }
}

Plan: 0 to add, 0 to change, 48 to destroy.
  
```


The screenshot shows the GitHub Actions interface for a workflow named '10-Terraform Destroy #21'. The workflow is in a 'Completed' state, indicated by a green checkmark. The left sidebar shows the workflow steps: 'Terraform Plan', 'Terraform Apply', and 'Backend Destroy'. The 'Terraform Apply' job is selected, and its details are shown on the right. The job 'Terraform Apply' succeeded 19 hours ago in 8m 38s. The job steps are: 'Set up job', 'Checkout', 'Setup Terraform', 'Terraform Init', 'Download Terraform Plan', and 'Terraform Apply'. The 'Terraform Apply' step is expanded, showing the command 'Run terraform apply -auto-approve tfplandestroy/tfplandestroy' and the output logs, which include the destruction of various Kubernetes resources.

10-Terraform Destroy #21

Summary

Jobs

- Terraform Plan
- Terraform Apply
- Backend Destroy

Run details

- Usage
- Workflow file

Terraform Apply

succeeded 19 hours ago in 8m 38s

- Set up job
- Checkout
- Setup Terraform
- Terraform Init
- Download Terraform Plan
- Terraform Apply

```
1 ▶ Run terraform apply -auto-approve tfplandestroy/tfplandestroy
19 module.application.kubernetes.horizontal_pod_autoscaler.hpa[jls]: Destroying... [id-dev-apps-jeanphilipps/k8s-hpa-dev-jeanphilipps]
20 module.sec.helm.release.sonarqube: Destroying... [id-sonarqdevjpls]
21 module.application.kubernetes.horizontal_pod_autoscaler.hpa[jls]: Destruction complete after 1s
22 module.application.kubernetes.deployment.apps[k8s]: Destroying... [id-dev-apps-jeanphilipps/devops-deploy-jeanphilipps]
23 module.application.kubernetes.deployment.apps[k8s]: Destruction complete after 0s
24 module.application.kubernetes.cluster_role_binding.allow_deploy: Destroying... [id-CR-BIND-dev-DEPLOY]
25 module.application.kubernetes.service.apps[k8s]: Destroying... [id-dev-apps-jeanphilipps/devopsdevpods]
26 module.application.kubernetes.cluster_role_binding.allow_deploy: Destruction complete after 0s
```

The screenshot shows the GitHub Actions interface for the same workflow '10-Terraform Destroy #21'. The 'Backend Destroy' job is selected, and its details are shown on the right. The job 'Backend Destroy' succeeded 19 hours ago in 1m 27s. The job steps are: 'Set up job', 'Checkout', 'Destroy Backend on Azure', 'Post Checkout', and 'Complete job'.

10-Terraform Destroy #21

Summary

Jobs

- Terraform Plan
- Terraform Apply
- Backend Destroy

Run details

- Usage
- Workflow file

Backend Destroy

succeeded 19 hours ago in 1m 27s

- Set up job
- Checkout
- Destroy Backend on Azure
- Post Checkout
- Complete job